

[MC-IISA]:

Internet Information Services (IIS) Application Host COM Protocol

Intellectual Property Rights Notice for Open Specifications Documentation

- **Technical Documentation.** Microsoft publishes Open Specifications documentation ("this documentation") for protocols, file formats, data portability, computer languages, and standards support. Additionally, overview documents cover inter-protocol relationships and interactions.
- **Copyrights.** This documentation is covered by Microsoft copyrights. Regardless of any other terms that are contained in the terms of use for the Microsoft website that hosts this documentation, you can make copies of it in order to develop implementations of the technologies that are described in this documentation and can distribute portions of it in your implementations that use these technologies or in your documentation as necessary to properly document the implementation. You can also distribute in your implementation, with or without modification, any schemas, IDLs, or code samples that are included in the documentation. This permission also applies to any documents that are referenced in the Open Specifications documentation.
- **No Trade Secrets.** Microsoft does not claim any trade secret rights in this documentation.
- **Patents.** Microsoft has patents that might cover your implementations of the technologies described in the Open Specifications documentation. Neither this notice nor Microsoft's delivery of this documentation grants any licenses under those patents or any other Microsoft patents. However, a given Open Specifications document might be covered by the Microsoft [Open Specifications Promise](#) or the [Microsoft Community Promise](#). If you would prefer a written license, or if the technologies described in this documentation are not covered by the Open Specifications Promise or Community Promise, as applicable, patent licenses are available by contacting iplg@microsoft.com.
- **License Programs.** To see all of the protocols in scope under a specific license program and the associated patents, visit the [Patent Map](#).
- **Trademarks.** The names of companies and products contained in this documentation might be covered by trademarks or similar intellectual property rights. This notice does not grant any licenses under those rights. For a list of Microsoft trademarks, visit www.microsoft.com/trademarks.
- **Fictitious Names.** The example companies, organizations, products, domain names, email addresses, logos, people, places, and events that are depicted in this documentation are fictitious. No association with any real company, organization, product, domain name, email address, logo, person, place, or event is intended or should be inferred.

Reservation of Rights. All other rights are reserved, and this notice does not grant any rights other than as specifically described above, whether by implication, estoppel, or otherwise.

Tools. The Open Specifications documentation does not require the use of Microsoft programming tools or programming environments in order for you to develop an implementation. If you have access to Microsoft programming tools and environments, you are free to take advantage of them. Certain Open Specifications documents are intended for use in conjunction with publicly available standards specifications and network programming art and, as such, assume that the reader either is familiar with the aforementioned material or has immediate access to it.

Support. For questions and support, please contact dochelp@microsoft.com.

Revision Summary

Date	Revision History	Revision Class	Comments
4/3/2007	0.1	Major	Initial Availability
7/3/2007	0.2	Minor	Clarified the meaning of the technical content.
8/10/2007	0.2.1	Editorial	Changed language and formatting in the technical content.
9/28/2007	0.3	Minor	Clarified the meaning of the technical content.
10/23/2007	0.4	Minor	Clarified the meaning of the technical content.
11/30/2007	0.5	Minor	Clarified the meaning of the technical content.
1/25/2008	0.5.1	Editorial	Changed language and formatting in the technical content.
3/14/2008	1.0	Major	Updated and revised the technical content.
5/16/2008	1.0.1	Editorial	Changed language and formatting in the technical content.
6/20/2008	2.0	Major	Updated and revised the technical content.
7/25/2008	2.0.1	Editorial	Changed language and formatting in the technical content.
8/29/2008	2.0.2	Editorial	Changed language and formatting in the technical content.
10/24/2008	2.0.3	Editorial	Changed language and formatting in the technical content.
12/5/2008	2.1	Minor	Clarified the meaning of the technical content.
1/16/2009	2.1.1	Editorial	Changed language and formatting in the technical content.
2/27/2009	2.1.2	Editorial	Changed language and formatting in the technical content.
4/10/2009	2.1.3	Editorial	Changed language and formatting in the technical content.
5/22/2009	2.1.4	Editorial	Changed language and formatting in the technical content.
7/2/2009	2.2	Minor	Clarified the meaning of the technical content.
8/14/2009	2.2.1	Editorial	Changed language and formatting in the technical content.
9/25/2009	2.3	Minor	Clarified the meaning of the technical content.
11/6/2009	2.3.1	Editorial	Changed language and formatting in the technical content.
12/18/2009	2.3.2	Editorial	Changed language and formatting in the technical content.
1/29/2010	2.3.3	Editorial	Changed language and formatting in the technical content.
3/12/2010	2.3.4	Editorial	Changed language and formatting in the technical content.
4/23/2010	3.0	Major	Updated and revised the technical content.
6/4/2010	4.0	Major	Updated and revised the technical content.
7/16/2010	5.0	Major	Updated and revised the technical content.
8/27/2010	5.0	None	No changes to the meaning, language, or formatting of the technical content.
10/8/2010	6.0	Major	Updated and revised the technical content.

Date	Revision History	Revision Class	Comments
11/19/2010	6.0	None	No changes to the meaning, language, or formatting of the technical content.
1/7/2011	6.0	None	No changes to the meaning, language, or formatting of the technical content.
2/11/2011	6.0	None	No changes to the meaning, language, or formatting of the technical content.
3/25/2011	6.0	None	No changes to the meaning, language, or formatting of the technical content.
5/6/2011	6.0	None	No changes to the meaning, language, or formatting of the technical content.
6/17/2011	6.1	Minor	Clarified the meaning of the technical content.
9/23/2011	6.1	None	No changes to the meaning, language, or formatting of the technical content.
12/16/2011	7.0	Major	Updated and revised the technical content.
3/30/2012	7.0	None	No changes to the meaning, language, or formatting of the technical content.
7/12/2012	7.0	None	No changes to the meaning, language, or formatting of the technical content.
10/25/2012	8.0	Major	Updated and revised the technical content.
1/31/2013	8.0	None	No changes to the meaning, language, or formatting of the technical content.
8/8/2013	9.0	Major	Updated and revised the technical content.
11/14/2013	9.0	None	No changes to the meaning, language, or formatting of the technical content.
2/13/2014	9.0	None	No changes to the meaning, language, or formatting of the technical content.
5/15/2014	9.0	None	No changes to the meaning, language, or formatting of the technical content.
6/30/2015	10.0	Major	Significantly changed the technical content.
10/16/2015	10.0	None	No changes to the meaning, language, or formatting of the technical content.
7/14/2016	10.0	None	No changes to the meaning, language, or formatting of the technical content.
6/1/2017	10.0	None	No changes to the meaning, language, or formatting of the technical content.
9/15/2017	11.0	Major	Significantly changed the technical content.
9/12/2018	12.0	Major	Significantly changed the technical content.
4/7/2021	13.0	Major	Significantly changed the technical content.
6/25/2021	14.0	Major	Significantly changed the technical content.

Date	Revision History	Revision Class	Comments
4/23/2024	15.0	Major	Significantly changed the technical content.

Table of Contents

1	Introduction	10
1.1	Glossary	10
1.2	References	10
1.2.1	Normative References	11
1.2.2	Informative References	11
1.3	Overview	11
1.4	Relationship to Other Protocols	11
1.5	Prerequisites/Preconditions	11
1.6	Applicability Statement	11
1.7	Versioning and Capability Negotiation	11
1.8	Vendor-Extensible Fields	12
1.9	Standards Assignments.....	12
2	Messages.....	14
2.1	Transport	14
2.2	Common Data Types	14
3	Protocol Details	15
3.1	IIS Application Host Administration Server Details.....	15
3.1.1	Abstract Data Model.....	15
3.1.1.1	Configuration Store	15
3.1.1.2	Configuration Path	15
3.1.1.3	Configuration Settings	15
3.1.1.3.1	Configuration Properties	16
3.1.1.3.2	Configuration Elements	16
3.1.1.3.3	Configuration Collections	16
3.1.1.3.4	Configuration Sections and Section Definition and Section Groups	17
3.1.1.3.5	Configuration Schema	17
3.1.1.3.6	Inheritance and Merging of Configuration Settings.....	17
3.1.1.3.7	Programmatic Configuration Settings	18
3.1.2	Timers	18
3.1.3	Initialization.....	18
3.1.4	Message Processing Events and Sequencing Rules	18
3.1.4.1	IAppHostAdminManager.....	18
3.1.4.1.1	GetAdminSection (Opnum 3)	19
3.1.4.1.2	GetMetadata (Opnum 4).....	20
3.1.4.1.3	SetMetadata (Opnum 5).....	22
3.1.4.1.4	ConfigManager (Opnum 6).....	24
3.1.4.2	IAppHostChangeHandler	24
3.1.4.2.1	OnSectionChanges (Opnum 3)	25
3.1.4.3	IAppHostChildElementCollection.....	25
3.1.4.3.1	Count (Opnum 3)	25
3.1.4.3.2	Item (Opnum 4)	26
3.1.4.4	IAppHostCollectionSchema	27
3.1.4.4.1	AddElementNames (Opnum 3)	27
3.1.4.4.2	GetAddElementSchema (Opnum 4).....	28
3.1.4.4.3	RemoveElementSchema (Opnum 5).....	29
3.1.4.4.4	ClearElementSchema (Opnum 6).....	29
3.1.4.4.5	IsMergeAppend (Opnum 7).....	30
3.1.4.4.6	GetMetadata (Opnum 8).....	30
3.1.4.4.7	DoesAllowDuplicates (Opnum 9).....	30
3.1.4.5	IAppHostConfigException	31
3.1.4.5.1	LineNumber (Opnum 3).....	31
3.1.4.5.2	FileName (Opnum 4)	32
3.1.4.5.3	ConfigPath (Opnum 5)	32

3.1.4.5.4	ErrorLine (Opnum 6).....	33
3.1.4.5.5	PreErrorLine (Opnum 7)	33
3.1.4.5.6	PostErrorLine (Opnum 8).....	34
3.1.4.5.7	ErrorString (Opnum 9).....	35
3.1.4.6	IAppHostConfigFile.....	35
3.1.4.6.1	ConfigPath (Opnum 3)	36
3.1.4.6.2	FilePath (Opnum 4)	36
3.1.4.6.3	Locations (Opnum 5)	37
3.1.4.6.4	GetAdminSection (Opnum 6)	37
3.1.4.6.5	GetMetadata (Opnum 7).....	38
3.1.4.6.6	SetMetadata (Opnum 8).....	39
3.1.4.6.7	ClearInvalidSections (Opnum 9).....	39
3.1.4.6.8	RootSectionGroup (Opnum 10)	39
3.1.4.7	IAppHostConfigLocation.....	40
3.1.4.7.1	Path (Opnum 3)	41
3.1.4.7.2	Count (Opnum 4)	41
3.1.4.7.3	Item (Opnum 5).....	42
3.1.4.7.4	AddConfigSection (Opnum 6)	42
3.1.4.7.5	DeleteConfigSection (Opnum 7)	43
3.1.4.8	IAppHostConfigLocationCollection	44
3.1.4.8.1	Count (Opnum 3)	44
3.1.4.8.2	Item (Opnum 4).....	45
3.1.4.8.3	AddLocation (Opnum 5)	45
3.1.4.8.4	DeleteLocation (Opnum 6).....	46
3.1.4.9	IAppHostConfigManager.....	47
3.1.4.9.1	GetConfigFile (Opnum 3).....	47
3.1.4.9.2	GetUniqueConfigPath (Opnum 4).....	48
3.1.4.10	IAppHostConstantValue	49
3.1.4.10.1	Name (Opnum 3)	49
3.1.4.10.2	Value (Opnum 4).....	50
3.1.4.11	IAppHostConstantValueCollection.....	50
3.1.4.11.1	Count (Opnum 3)	50
3.1.4.11.2	Item (Opnum 4).....	51
3.1.4.12	IAppHostElement	52
3.1.4.12.1	Name (Opnum 3)	52
3.1.4.12.2	Collection (Opnum 4).....	53
3.1.4.12.3	Properties (Opnum 5)	53
3.1.4.12.4	ChildElements (Opnum 6).....	54
3.1.4.12.5	GetMetadata (Opnum 7).....	55
3.1.4.12.6	SetMetadata (Opnum 8).....	57
3.1.4.12.7	Schema (Opnum 9)	58
3.1.4.12.8	GetElementByName (Opnum 10).....	59
3.1.4.12.9	GetPropertyByName (Opnum 11)	59
3.1.4.12.10	Clear (Opnum 12)	60
3.1.4.12.11	Methods (Opnum 13).....	60
3.1.4.13	IAppHostElementCollection.....	61
3.1.4.13.1	Count (Opnum 3)	62
3.1.4.13.2	Item (Opnum 4).....	62
3.1.4.13.3	AddElement (Opnum 5).....	63
3.1.4.13.4	DeleteElement (Opnum 6)	63
3.1.4.13.5	Clear (Opnum 7)	64
3.1.4.13.6	CreateNewElement (Opnum 8).....	65
3.1.4.13.7	Schema (Opnum 9)	65
3.1.4.14	IAppHostElementSchema	66
3.1.4.14.1	Name (Opnum 3)	66
3.1.4.14.2	DoesAllowUnschematizedProperties (Opnum 4)	67
3.1.4.14.3	GetMetadata (Opnum 5).....	67
3.1.4.14.4	CollectionSchema (Opnum 6).....	68

3.1.4.14.5	ChildElementSchemas (Opnum 7)	69
3.1.4.14.6	PropertySchemas (Opnum 8)	69
3.1.4.14.7	IsCollectionDefault (Opnum 9)	70
3.1.4.15	IAppHostElementSchemaCollection	70
3.1.4.15.1	Count (Opnum 3)	71
3.1.4.15.2	Item (Opnum 4)	71
3.1.4.16	IAppHostMappingExtension	72
3.1.4.16.1	GetSiteNameFromSiteID (Opnum 3)	72
3.1.4.16.2	GetSiteIDFromSiteName (Opnum 4)	73
3.1.4.16.3	GetSiteElementFromSiteID (Opnum 5)	74
3.1.4.16.4	MapPath (Opnum 6)	74
3.1.4.17	IAppHostMethod	75
3.1.4.17.1	Name (Opnum 3)	76
3.1.4.17.2	Schema (Opnum 4)	76
3.1.4.17.3	CreateInstance (Opnum 5)	77
3.1.4.18	IAppHostMethodCollection	77
3.1.4.18.1	Count (Opnum 3)	77
3.1.4.18.2	Item (Opnum 4)	78
3.1.4.19	IAppHostMethodInstance	79
3.1.4.19.1	Input (Opnum 3)	79
3.1.4.19.2	Output (Opnum 4)	79
3.1.4.19.3	Execute (Opnum 5)	80
3.1.4.19.4	GetMetadata (Opnum 6)	80
3.1.4.19.5	SetMetadata (Opnum 7)	81
3.1.4.20	IAppHostMethodSchema	82
3.1.4.20.1	Name (Opnum 3)	82
3.1.4.20.2	InputSchema (Opnum 4)	83
3.1.4.20.3	OutputSchema (Opnum 5)	83
3.1.4.20.4	GetMetadata (Opnum 6)	83
3.1.4.21	IAppHostPathMapper	84
3.1.4.21.1	MapPath (Opnum 3)	84
3.1.4.22	IAppHostProperty	85
3.1.4.22.1	Name (Opnum 3)	86
3.1.4.22.2	Value (Get) (Opnum 4)	86
3.1.4.22.3	Value (Set) (Opnum 5)	87
3.1.4.22.4	Clear (Opnum 6)	87
3.1.4.22.5	StringValue (Opnum 7)	88
3.1.4.22.6	Exception (Opnum 8)	88
3.1.4.22.7	GetMetadata (Opnum 9)	89
3.1.4.22.8	SetMetadata (Opnum 10)	90
3.1.4.22.9	Schema (Opnum 11)	91
3.1.4.23	IAppHostPropertyCollection	91
3.1.4.23.1	Count (Opnum 3)	92
3.1.4.23.2	Item (Opnum 4)	92
3.1.4.24	IAppHostPropertyException	93
3.1.4.24.1	InvalidValue (Opnum 10)	93
3.1.4.24.2	ValidationFailureReason (Opnum 11)	94
3.1.4.24.3	ValidationFailureParameters (Opnum 12)	94
3.1.4.25	IAppHostPropertySchema	95
3.1.4.25.1	Name (Opnum 3)	95
3.1.4.25.2	Type (Opnum 4)	96
3.1.4.25.3	DefaultValue (Opnum 5)	96
3.1.4.25.4	IsRequired (Opnum 6)	97
3.1.4.25.5	IsUniqueKey (Opnum 7)	97
3.1.4.25.6	IsCombinedKey (Opnum 8)	98
3.1.4.25.7	IsExpanded (Opnum 9)	98
3.1.4.25.8	ValidationType (Opnum 10)	99
3.1.4.25.9	ValidationParameter (Opnum 11)	99

3.1.4.25.10	GetMetaData (Opnum 12)	100
3.1.4.25.11	IsCaseSensitive (Opnum 13).....	101
3.1.4.25.12	PossibleValues (Opnum 14)	101
3.1.4.25.13	DoesAllowInfinite (Opnum 15)	102
3.1.4.25.14	IsEncrypted (Opnum 16)	102
3.1.4.25.15	TimeSpanFormat (Opnum 17).....	103
3.1.4.26	IAppHostPropertySchemaCollection	103
3.1.4.26.1	Count (Opnum 3)	104
3.1.4.26.2	Item (Opnum 4).....	104
3.1.4.27	IAppHostSectionDefinition	105
3.1.4.27.1	Name (Opnum 3)	105
3.1.4.27.2	Type (Get) (Opnum 4)	106
3.1.4.27.3	Type (Set) (Opnum 5)	106
3.1.4.27.4	OverrideModeDefault (Get) (Opnum 6)	107
3.1.4.27.5	OverrideModeDefault (Set) (Opnum 7).....	108
3.1.4.27.6	AllowDefinition (Get) (Opnum 8)	108
3.1.4.27.7	AllowDefinition (Set) (Opnum 9)	109
3.1.4.27.8	AllowLocation (Get) (Opnum 10)	109
3.1.4.27.9	AllowLocation (Set) (Opnum 11)	110
3.1.4.28	IAppHostSectionDefinitionCollection	110
3.1.4.28.1	Count (Opnum 3)	111
3.1.4.28.2	Item (Opnum 4).....	111
3.1.4.28.3	AddSection (Opnum 5).....	112
3.1.4.28.4	DeleteSection (Opnum 6)	112
3.1.4.29	IAppHostSectionGroup.....	113
3.1.4.29.1	Count (Opnum 3)	113
3.1.4.29.2	Item (Opnum 4).....	114
3.1.4.29.3	Sections (Opnum 5).....	115
3.1.4.29.4	AddSectionGroup (Opnum 6)	115
3.1.4.29.5	DeleteSectionGroup (Opnum 7).....	116
3.1.4.29.6	Name (Opnum 8)	116
3.1.4.29.7	Type (Get) (Opnum 9)	117
3.1.4.29.8	Type (Set) (Opnum 10).....	117
3.1.4.30	IAppHostWritableAdminManager	118
3.1.4.30.1	CommitChanges (Opnum 7).....	118
3.1.4.30.2	CommitPath (Get) (Opnum 8).....	119
3.1.4.30.3	CommitPath (Set) (Opnum 9)	119
3.1.5	Timer Events.....	120
3.1.6	Other Local Events.....	120
4	Protocol Examples	121
4.1	Create an AppHostAdminManager Locally	121
4.2	Get Metadata: Get the overrideMode of a defaultDocument Section.....	121
4.3	Set Metadata: Set the overrideMode of the defaultDocument Section	121
4.4	Create a New Configuration Section Entry in the configSections Section	121
4.5	Get a Section for Read Access: The defaultDocument Section	122
4.6	Get a Property: Get the Enabled Property of the defaultDocument Section	122
4.7	Get a Section: Get the anonymousAuthentication Section	122
4.8	List the Entries of a Collection.....	122
4.9	Remove an Entry of a Collection	122
4.10	Edit the Configuration of APPHOST in a Location Tag	123
4.11	Read Schema Information: Determine If IsMergeAppend Is Set in the defaultDocuments Section	123
4.12	Get a Section for Write: Get the defaultDocument Section and Toggle the Enabled Attribute	123
4.13	Write into a Collection: Clear the Contents of the defaultDocument Section for Site1	123
4.14	Write into a Collection: Add an Entry for the defaultDocument Section for Site1 as a Location Tag.....	124

5	Security	125
5.1	Security Considerations for Implementers	125
5.2	Index of Security Parameters	125
6	Appendix A: Full IDL.....	126
7	Appendix B: Product Behavior	140
8	Change Tracking.....	141
9	Index.....	142

1 Introduction

This document specifies the Internet Information Services (IIS) Application Host COM Protocol. This protocol is a client-to-server protocol that enables remote read/write access to server data. The server data can be used to define administration, configuration, and operational parameters to an application server service, which can be a web server.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

class identifier (CLSID): A GUID that identifies a software component; for instance, a DCOM object class or a COM class.

dynamic endpoint: A network-specific server address that is requested and assigned at run time. For more information, see [\[C706\]](#).

interface: A specification in a Component Object Model (COM) server that describes how to access the methods of a class. For more information, see [\[MS-DCOM\]](#).

Interface Definition Language (IDL): The International Standards Organization (ISO) standard language for specifying the **interface** for remote procedure calls. For more information, see [\[C706\]](#) section 4.

opnum: An operation number or numeric identifier that is used to identify a specific **remote procedure call (RPC)** method or a method in an interface. For more information, see [\[C706\]](#) section 12.5.2.12 or [\[MS-RPCE\]](#).

remote procedure call (RPC): A communication protocol used primarily between client and server. The term has three definitions that are often used interchangeably: a runtime environment providing for communication facilities between computers (the RPC runtime); a set of request-and-response message exchanges between computers (the RPC exchange); and the single message from an RPC exchange (the RPC message). For more information, see [\[C706\]](#).

universally unique identifier (UUID): A 128-bit value. UUIDs can be used for multiple purposes, from tagging objects with an extremely short lifetime, to reliably identifying very persistent objects in cross-process communication such as client and server interfaces, manager entry-point vectors, and **RPC** objects. UUIDs are highly likely to be unique. UUIDs are also known as globally unique identifiers (GUIDs) and these terms are used interchangeably in the Microsoft protocol technical documents (TDs). Interchanging the usage of these terms does not imply or require a specific algorithm or mechanism to generate the UUID. Specifically, the use of this term does not imply or require that the algorithms described in [\[RFC4122\]](#) or [\[C706\]](#) has to be used for generating the UUID.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[C706] The Open Group, "DCE 1.1: Remote Procedure Call", C706, August 1997, <https://publications.opengroup.org/c706>

Note Registration is required to download the document.

[MS-DCOM] Microsoft Corporation, "[Distributed Component Object Model \(DCOM\) Remote Protocol](#)".

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-ERREF] Microsoft Corporation, "[Windows Error Codes](#)".

[MS-OAUT] Microsoft Corporation, "[OLE Automation Protocol](#)".

[MS-RPCE] Microsoft Corporation, "[Remote Procedure Call Protocol Extensions](#)".

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

1.2.2 Informative References

[MSDN-IIS7AH] Microsoft Corporation, "IIS 7.0: IIS Application Host Administration API Reference", <http://msdn.microsoft.com/en-us/library/aa965120.aspx>

1.3 Overview

This protocol is intended to provide read/write access to administrative configuration data that is located on a remote server computer. The administrative configuration data is implementation-specific for each server.

1.4 Relationship to Other Protocols

This protocol depends on the protocol described in [\[MS-DCOM\]](#).

1.5 Prerequisites/Preconditions

This protocol requires that the protocol specified in [\[MS-DCOM\]](#) is implemented on both the client and server computers.

This protocol specification assumes that any security or authentication associations between the client and server be performed by the DCOM layer.

1.6 Applicability Statement

This protocol is applicable to the remote reading and writing of hierarchically organized server administration data.

1.7 Versioning and Capability Negotiation

This protocol does not provide a mechanism for protocol versioning or capability negotiation.

1.8 Vendor-Extensible Fields

This protocol does not include any vendor-extensible fields.

This protocol uses HRESULTs that are vendor-extensible, as specified in [MS-ERREF] section 2.1. Vendors can choose their own values for this field as long as the C bit (0x20000000) is set, indicating it is a customer code.

This protocol uses Win32 error codes. These values are taken from the numbering space of the Windows error codes, as specified in [MS-ERREF] section 2.2. Vendors SHOULD use those values, retaining their default meaning. Choosing any other meaning for these values risks a future collision.

1.9 Standards Assignments

The following parameters are implementation-specific proprietary assignments. <1>.

Parameter	Value	Reference
DCOM CLSID for the AppHostAdminManager	228fb8f7-fb53-4fd5-8c7b-ff59de606c5b	None
DCOM CLSID for the AppHostWritableAdminManager	2b72133b-3f5b-4602-8952-803546ce3344	None
RPC Interface UUID for IAppHostMappingExtension	31a83ea0-c0e4-4a2c-8a01-353cc2a4c60a	None
RPC Interface UUID for IAppHostChildElementCollection	08a90f5f-0702-48d6-b45f-02a9885a9768	None
RPC Interface UUID for IAppHostPropertyCollection	0191775e-bcff-445a-b4f4-3bdda54e2816	None
RPC Interface UUID for IAppHostConfigLocationCollection	832a32f7-b3ea-4b8c-b260-9a2923001184	None
RPC Interface UUID for IAppHostMethodCollection	d6c7cd8f-bb8d-4f96-b591-d3a5f1320269	None
RPC Interface UUID for IAppHostElementSchemaCollection	0344cdda-151e-4cbf-82da-66ae61e97754	None
RPC Interface UUID for IAppHostPropertySchemaCollection	8bed2c68-a5fb-4b28-8581-a0dc5267419f	None
RPC Interface UUID for IAppHostConstantValueCollection	5b5a68e6-8b9f-45e1-8199-a95ffccdffff	None
RPC Interface UUID for IAppHostConstantValue	0716caf8-7d05-4a46-8099-77594be91394	None
RPC Interface UUID for IAppHostPropertySchema	450386db-7409-4667-935e-384dbbee2a9e	None
RPC Interface UUID for IAppHostCollectionSchema	de095db1-5368-4d11-81f6-efef619b7bcf	None
RPC Interface UUID for IAppHostElementSchema	ef13d885-642c-4709-99ec-b89561c6bc69	None
RPC Interface UUID for IAppHostMethodSchema	2d9915fb-9d42-4328-b782-1b46819fab9e	None
RPC Interface UUID for IAppHostMethodInstance	b80f3c42-60e0-4ae0-9007-f52852d3dbed	None

Parameter	Value	Reference
RPC Interface UUID for IAppHostMethod	7883ca1c-1112-4447-84c3-52fbeb38069d	None
RPC Interface UUID for IAppHostConfigException	4dfa1df3-8900-4bc7-bbb5-d1a458c52410	None
RPC Interface UUID for IAppHostPropertyException	eafe4895-a929-41ea-b14d-613e23f62b71	None
RPC Interface UUID for IAppHostElementCollection	c8550bff-5281-4b1e-ac34-99b6fa38464d	None
RPC Interface UUID for IAppHostElement	64ff8ccc-b287-4dae-b08a-a72cbf45f453	None
RPC Interface UUID for IAppHostProperty	ed35f7a1-5024-4e7b-a44d-07ddaf4b524d	None
RPC Interface UUID for IAppHostConfigLocation	370af178-7758-4dad-8146-7391f6e18585	None
RPC Interface UUID for IAppHostSectionDefinition	c5c04795-321c-4014-8fd6-d44658799393	None
RPC Interface UUID for IAppHostSectionDefinitionCollection	b7d381ee-8860-47a1-8af4-1f33b2b1f325	None
RPC Interface UUID for IAppHostSectionGroup	0dd8a158-ebe6-4008-a1d9-b7ecc8f1104b	None
RPC Interface UUID for IAppHostConfigFile	ada4e6fb-e025-401e-a5d0-c3134a281f07	None
RPC Interface UUID for IAppHostPathMapper	e7927575-5cc3-403b-822e-328a6b904bee	None
RPC Interface UUID for IAppHostChangeHandler	09829352-87c2-418d-8d79-4133969a489d	None
RPC Interface UUID for IAppHostAdminManager	9be77978-73ed-4a9a-87fd-13f09fec1b13	None
RPC Interface UUID for IAppHostWritableAdminManager	fa7660f6-7b3f-4237-a8bf-ed0ad0dcbbd9	None
RPC Interface UUID for IAppHostConfigManager	8f6d760f-f0cb-4d69-b5f6-848b33e9bdc6	None

2 Messages

2.1 Transport

This protocol MUST use the DCOM Remote Protocol, as specified in [\[MS-DCOM\]](#), as its transport. On its behalf, the DCOM Remote Protocol uses the following **remote procedure call (RPC)** protocol sequence: RPC over TCP, as specified in [\[MS-RPCE\]](#). This protocol uses RPC **dynamic endpoints**, as specified in [\[C706\]](#) section 4.

This protocol MUST use the RPC **UUIDs** specified in section [1.9](#).

To receive incoming remote calls for these **interfaces**, the server MUST implement a DCOM Object Class with the **CLSIDs** (specified in section 1.9) [AppHostAdminManager](#) using the UUID {228fb8f7-fb53-4fd5-8c7b-ff59de606c5b}, and [AppHostWritableAdminManager](#) using the UUID {2b72133b-3f5b-4602-8952-803546ce3344}.

2.2 Common Data Types

None.

3 Protocol Details

The client side of this protocol is simply a pass-through. That is, no additional timers or other states are required on the client side of this protocol. Calls that are made by any client application are passed directly to the transport, and the results that are returned by the transport are passed directly back to the calling application.

3.1 IIS Application Host Administration Server Details

3.1.1 Abstract Data Model

This section describes a conceptual data model that an implementation **MUST** expose to participate in this protocol. The model is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that described in this document.

3.1.1.1 Configuration Store

This protocol is used to manage administrative data that is organized in a hierarchical manner. Each node in the hierarchy is identified by a name. Node names are strings that cannot contain the '/' character. A node can also have one or more child nodes. Node names **MUST** be unique across sibling nodes. Every node is uniquely identified using a fully qualified name known as a configuration path (see section [3.1.1.2](#)).

Each node of the configuration store can store configuration settings (see section [3.1.1.3](#)).

3.1.1.2 Configuration Path

Each node can be uniquely identified by a fully qualified name. A fully qualified name consists of names of all the ancestor nodes listed in order beginning with the root node of the tree. Node names in the fully qualified path are separated by the '/' character.

A fully qualified path that identifies a node in the [configuration store](#) is called the configuration path.

For the following example of configuration nodes:

```
Machine
  Webroot
    AppHost
      Site1
        App1
          Dir1
          Dir2
        App2
      Site2
        App3
        App4
```

Node **Machine** has a single child node **Webroot**. Node **Site1** has a child node called **App1** that in turn has two child nodes **Dir1** and **Dir2**. The configuration path for node **App1** is **Machine/Webroot/Apphost/Site1/App1**.

3.1.1.3 Configuration Settings

Configuration settings are where the actual configuration data is stored. Configuration settings are organized in one of three ways, [configuration properties](#), [configuration elements](#), or [configuration collections](#). Configuration data is ultimately stored in configuration properties. Configuration elements and configuration collections are used to organized and group together different configuration properties in a logical way.

3.1.1.3.1 Configuration Properties

A configuration property has a name and can store a primitive value of a specific set of types including **boolean**, **integer**, **string**, and other standard types. The configuration property name **MUST** be unique in relation to its parent [configuration element](#). A configuration property cannot have any children.

3.1.1.3.2 Configuration Elements

A configuration element is designed to help with the logical organization of data. It can contain zero, or more configuration elements, [configuration properties](#), and [configuration collections](#).

It provides the means to organize and group together in a functional way a set of otherwise flat configuration properties.

As an example, assume there are anonymous authentication and Windows authentication features that need to store settings. The <anonymousAuthentication> element could store all the settings for anonymous authentication features while the <windowsAuthentication> element could store all the settings for Windows authentication features.

To further organize the settings, the <anonymousAuthentication> and <windowsAuthentication> elements could be grouped together under a configuration element called <authentication>. The <authentication> element can be organized under a parent element called <security>.

```
system.webServer
  security
    authentication
      anonymousAuthentication
        enabled="true"
      windowsAuthentication
        enabled="false"
```

Fully qualified name for the <anonymousAuthentication> element within the hierarchy of configuration elements would be **system.webServer/security/authentication/anonymousAuthentication**.

3.1.1.3.3 Configuration Collections

Configuration collections are special [configuration elements](#) that serve as containers, where elements inside them are considered to be part of a set that can be dynamically added and removed. Elements in these containers are also known as collection items. A configuration item can also contain child elements and [configuration properties](#). The items contained in a configuration collection don't have to be uniquely named and they share the same schema (see section [3.1.1.3.5](#)).

Configuration items as members of configuration collections are structurally the same as configuration elements but semantically represent operation on the collection. These operations include such as add element, remove element, or clear the collection.

- The name of the collection item is the directive. By default, the directives are **add**, **remove**, and **clear**. However the configuration schema allows these names to be overridden for a collection. It implies from the above that collection item names are not unique within a collection.

- The configuration properties in a collection item describe additional data consumed by the operation.

Configuration schema can be used to establish uniqueness and other constraints on a configuration collection.

3.1.1.3.4 Configuration Sections and Section Definition and Section Groups

A **configuration section** is a [configuration element](#) that contains logically related [configuration settings](#). A **configuration section** is typically consumed by a functional component on the server, such as an authentication module on a web server. **Configuration sections** cannot be nested.

In order to be able to use a **configuration section**, the section must be defined either in the current [configuration path](#) where it will be used or on a higher or parent configuration path in the configuration hierarchy. This definition is called a **section definition**.

Configuration sections can be optionally grouped together into **section groups**.

Every **configuration section** has a name. The short name is the name of the **configuration section** itself. The long name is the full name including all the containing **section groups**. For example, the full name for the configuration section called "windowsAuthentication" could be "system.webServer/security/authentication/windowsAuthentication" where system.webServer, security, and authentication would be configuration **section groups**. This hierarchical organization allows **configuration sections** and **section groups** with the same name, but under different **section groups**.

3.1.1.3.5 Configuration Schema

The structure of the [configuration settings](#) is described using a schema. A client can use configuration schema to discover, navigate and manipulate the [configuration store](#).

The schema is the definition of the configuration settings including its layout, validation and additional metadata.

The configuration schema contains complete information about [configuration elements](#), [configuration collections](#), and [configuration properties](#). Each configuration property is described by its type, default value, and validation criteria. The configuration schema for each configuration collection includes information about the directive name of the element used for adding and removing collection elements. The default directive for adding is **add**, for removing is **remove**, and for clearing, which deletes all the values inherited from an ancestor, is **clear**. The schema also includes a description for merging semantics to detail, if configuration elements inherited from ancestor are appended after ancestor elements or pre-pended before ancestor elements (see section [3.1.1.3.6](#)).

3.1.1.3.6 Inheritance and Merging of Configuration Settings

The configuration system supports [configuration settings](#) inheritance along the [configuration path](#).

[Configuration properties](#) defined at ancestral configuration paths will be automatically inherited to child paths. The value of configuration property that is defined at the ancestral configuration path closest to the requested configuration path will be used. If a configuration property of [configuration element](#) is not defined on the current configuration path level nor defined anywhere on any of the ancestral paths, then the default value from the [configuration schema](#) will be used.

The configuration system also supports [configuration collection](#) inheritance and merging along the configuration path. Configuration collection elements from ancestral configuration paths are inherited to child configuration paths. New collection elements can be added on child paths. Existing collection elements inherited from parent nodes can be deleted. All collection elements inherited from a parent can be deleted by using the **clear** collection element. The configuration schema doesn't support

default collections. If collection is not configured directly on the requested configuration path nor on any of the parent configuration paths, it will be empty.

The exact merging rules are as follows:

- First, the configuration system will check schema to determine if collection supports append or prepend semantics.
 - If it is append, then collection elements from child node will always be listed after its parent's collection elements.
 - In the case of prepend merge, the collection elements from child node will be listed before its parent's collection elements.
- The configuration system will walk the full configuration path from the root down.
 - For each path, it will read locally defined collection elements.
 - It will clear all the elements from parent if a **clear** directive is detected.
 - It will remove all the elements flagged with the **remove** directive.
 - It will add elements of **add** type to the end of the element list in the case of append merge, or to the beginning of the list in the case of prepend merge.
 - It will fail if it detects a duplicate entry. The configuration schema contains information about which configuration property or set of properties are to be unique.

A merged configuration element for a given configuration path is a composite of all configuration properties and collection elements defined on the given configuration path, parent paths, or the default value in the configuration schema. These inheritance and merging rules are used for calculation of merged configuration elements.

3.1.1.3.7 Programmatic Configuration Settings

The configuration system MAY support custom server-side extensions. This allows some of the [configuration settings](#) to be generated programmatically as opposed to being expressed in storage. The IIS Application Host COM protocol allows clients to query if a [configuration element](#) or [configuration property](#) is implemented by a server-side extension. Beyond that, server-side extensions are an implementation detail and not governed by this specification.

3.1.2 Timers

This protocol implementation does not require the explicit use of any timers.

3.1.3 Initialization

This protocol uses DCOM initialization.

3.1.4 Message Processing Events and Sequencing Rules

In the **interfaces** that are described in the following sections, no exceptions are thrown except those that are thrown by the underlying **RPC** protocol, as specified in [\[MS-RPCE\]](#).

3.1.4.1 IAppHostAdminManager

The IAppHostAdminManager interface represents a read-only **interface** to an administration system that is implemented on the server.

The administration system consists primarily of a set of administration objects of varying complexity, which are accessed through the [IAppHostElement](#) interface (section 3.1.4.12) and exist at one or more paths that are exposed by the server. The administration system allows access to individual IAppHostElement objects that are available at specific paths and also provides access to merged IAppHostElement objects that consist of the merged contents of individual IAppHostElement objects.

Secondarily, the administration system provides access to, and the setting of, specific system behaviors that are available for administration objects, which are represented by the term "metadata". Metadata allows a caller to modify and inspect the behavior of the administration system.

The IAppHostAdminManager interface is used by clients to access (for read-only purposes) the contents of the administration system without altering the contents of the system. A tool that seeks to display the administration objects that are contained in the system is an example of a consumer of the IAppHostAdminManager interface.

The IAppHostAdminManager interface inherits **opnums** 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
GetAdminSection	Opnum: 3
GetMetadata	Opnum: 4
SetMetadata	Opnum: 5
ConfigManager	Opnum: 6

3.1.4.1.1 GetAdminSection (Opnum 3)

The GetAdminSection method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostElement](#) that contains a merging of one or more individual IAppHostElement objects, which are set at one or more places in the path hierarchy that is specified in the method call.

```
HRESULT GetAdminSection(  
    [in] BSTR bstrSectionName,  
    [in] BSTR bstrPath,  
    [out, retval] IAppHostElement** ppAdminSection  
);
```

bstrSectionName: The name of the administration section to fetch. The server contains named IAppHostElement objects that are specified at one or more paths. The name of the IAppHostElement is called the "section name" of the element. This parameter represents the required section name. The section name syntax is specific to the implementation of the server.

bstrPath: The path hierarchy for which to find and merge IAppHostElement objects. The server uses this path as an indication of where to look for IAppHostElement objects. The server finds all the IAppHostElement objects that exist anywhere in this path. How the server parses this path and potentially maps it to the location of IAppHostElement objects is specific to each implementation.

ppAdminSection: Returns a merged IAppHostElement object that contains the merged contents of one or more IAppHostElement objects that are located at points in the hierarchy that is specified in *bstrPath*.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.
0X80070002 ERROR_FILE_NOT_FOUND	The server resource (for example, a file or database) corresponding to the path bstrPath could not be found.
0X80070005 ERROR_ACCESS_DENIED	Access to the server resource (for example, a file or database) corresponding to the path bstrPath was denied.
0X00000002 ERROR_PATH_NOT_FOUND	The section specified by bstrSectionName is not supported.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.1.2 GetMetadata (Opnum 4)

The GetMetadata method is received by the server in an RPC_REQUEST packet. The purpose of the method is to retrieve settings and other behavior-modifying attributes of the administration system. The administration system supports modifications through these settings and attributes. The metadata returns the details of these modifications to the caller. The returned metadata is referenced by a string name, and is thus extensible without changing the main Interface Definition Language (IDL) file. As administration systems implementers add more options to their systems, they can expose these options as named metadata through this method.

This method is used to get a metadata property.

```
HRESULT GetMetadata(
    [in] BSTR bstrMetadataType,
    [out, retval] VARIANT* pValue
);
```

bstrMetadataType: The name of the metadata property to fetch. Valid names are as follows.

Value	Meaning
"pathMapper"	Returns the IAppHostPathMapper object, which is a user-provided object that controls how the administration system implementation maps a specified path hierarchy to actual resources.
"pathMapper2"	Returns the IAppHostPathMapper2 object, which is a user-provided object that controls how the administration system implementation maps a specified path hierarchy to actual resources.

Value	Meaning
"changeHandler"	Returns the IAppHostChangeHandler object, which is a user-provided object that receives notification when the administration system detects that a part of the path hierarchy has changed.
"ignoreInvalidAttributes"	Returns the flag that controls whether the administration system ignores certain types of errors when evaluating the IAppHostProperty part of an IAppHostElement . If the flag is true, the administration system ignores these errors. If the flag is false, the administration system treats these IAppHostProperty errors as fatal and invalidates the container IAppHostElement. bool
"ignoreInvalidRanges"	Returns the flag that controls whether the administration system will ignore certain types of errors when evaluating the IAppHostProperty part of an IAppHostElement. If the flag is true, the administration system ignores these errors. If the flag is false, the administration system treats these IAppHostProperty errors as fatal and invalidates the container IAppHostElement. bool
"ignoreInvalidDecryption"	Returns the flag that controls whether the administration system will ignore decryption errors when evaluating the IAppHostProperty part of an IAppHostElement. If the flag is true, the administration system ignores these errors. If the flag is false, the administration system treats these IAppHostProperty errors as fatal and invalidates the container IAppHostElement. bool
"expandEnvironmentStrings"	Returns the flag that controls whether the administration system will expand environment variables when parsing the configuration system. If the flag is true, the administration system expands the variables. If the flag is false, the administration system does not expand the variables. bool
"disableExtensions"	Returns a flag that determines whether the administration system supports custom server-side extensions that can extend how the system evaluates IAppHostElement objects. If true, the system will not support custom server-side extensions. If false, the system will support custom server-side extensions. bool
"availableSections"	Returns a comma-separated series of strings that contains all the names of the available administration section names that are supported by the administration system. Section names represent the names of IAppHostElement objects. If the administration system supports an IAppHostElement that is named "X", then X must appear in the comma-delimited list that is returned. string
"mappingExtension"	Returns an object that is used to directly access the hierarchy mapping system of the administration system, and is accessed through the IAppHostMappingExtension interface . IAppHostMappingExtension
"hideExceptionPhysicalPath"	Returns the flag that controls whether the administration system will give information about the physical path of the configuration file in exception messages. If the flag is true, the administration system hides the physical path. If the flag is false, the administration system returns the physical path. bool

Value	Meaning
"lockMetadata"	Returns the flag that controls whether metadata can be modified with the SetMetadata call. If the flag is true, then further calls to SetMetadata will fail with a locking violation. If the flag is false, SetMetadata calls are allowed. bool

pValue: Returns the value of the specified metadata property, and the type depends on the property fetched. Upon success, the property is returned in a VARIANT, and the type of the value depends on the type of metadata that is requested (specified in bstrMetadataType).

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 E_OUTOFMEMORY	Not enough memory is available to process this command.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property specified by bstrMetadataType is not supported.

3.1.4.1.3 SetMetadata (Opnum 5)

The SetMetadata method is received by the server in an RPC_REQUEST packet. In response, the administration system implementation changes its behavior as specified by the provided metadata. This method is the opposite of the [GetMetadata](#) method, which retrieves a specified behavior property of the administration system. SetMetadata sets a specified behavior property of the administration system.

This method is used to set a metadata property.

```
HRESULT SetMetadata(
    [in] BSTR bstrMetadataType,
    [in] VARIANT value
);
```

bstrMetadataType: The name of the metadata property to set. Valid names are as follows.

Value	Meaning
"pathMapper"	Sets the IAppHostPathMapper object, which is a user-provided object that controls how the implementation of the administration system maps a specific path hierarchy to actual resources.
"pathMapper2"	Sets the IAppHostPathMapper2 object, which is a user-provided object that controls how the implementation of the administration system maps a specific

Value	Meaning
	path hierarchy to actual resources.
"changeHandler"	Sets the IAppHostChangeHandler object, which is a user-provided object that receives notification when the administration system detects that a part of the path hierarchy has changed.
"ignoreInvalidAttributes"	Sets the flag that controls whether the administration system will ignore certain types of errors when evaluating the IAppHostProperty part of an IAppHostElement . If the flag is true, the administration system ignores these errors. If the flag is false, the administration system treats these IAppHostProperty errors as fatal and invalidates the IAppHostElement container. bool
"ignoreInvalidRanges"	Sets the flag that controls whether the administration system will ignore certain range validation errors when it evaluates the IAppHostProperty part of an IAppHostElement. If the flag is true, the administration system ignores these errors. If the flag is false, the administration system treats these IAppHostProperty errors as fatal and invalidates the IAppHostElement container. bool
"ignoreInvalidDecryption"	Sets the flag that controls whether the administration system will ignore decryption errors when evaluating the IAppHostProperty part of an IAppHostElement. If the flag is true, the administration system ignores these errors. If the flag is false, the administration system treats these IAppHostProperty errors as fatal and invalidates the IAppHostElement container. bool
"expandEnvironmentStrings"	Sets the flag that controls whether the administration system will expand environment variables when parsing the configuration system. If the flag is true, the administration system expands the variables. If the flag is false, the administration system does not expand the variables. bool
"disableExtensions"	Sets a flag that determines whether the administration system supports custom server-side extensions that can extend how the system evaluates IAppHostElement objects. If true, the system will not support custom server-side extensions. If false, the system will support custom server-side extensions. bool
"hideExceptionPhysicalPath"	Sets the flag that controls whether the administration system will give information about the physical path of the configuration file in exception messages. If the flag is true, the administration system hides the physical path. If the flag is false, the administration system returns the physical path. bool
"lockMetadata"	Sets the flag that controls whether metadata can be modified with the SetMetadata call. If the flag is true, then further calls to SetMetadata will fail with a locking violation. If the flag is false, SetMetadata calls are allowed. bool

value: The value of the metadata property to set.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070021 ERROR_LOCK_VIOLATION	Metadata property cannot be set because the lockMetadata property is set to true.
0X80070032 ERROR_NOT_SUPPORTED	The request is not supported because bstrMetadataType is not one of the supported metadata properties.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.1.4 ConfigManager (Opnum 6)

The ConfigManager method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostConfigManager](#) that provides direct access to the supported path hierarchy of the administration system and access to individual [IAppHostElement](#) objects that are contained within.

```
[propget] HRESULT ConfigManager(
    [out, retval] IAppHostConfigManager** ppConfigManager
);
```

ppConfigManager: If the server successfully processes the message that is received from the client, *ppConfigManager* contains a pointer to an IAppHostConfigManager object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If successful, *ppConfigManager MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.2 IAppHostChangeHandler

The IAppHostChangeHandler describes an **interface** that clients can implement and that is called when the administration system has detected a change in a part of its path hierarchy.

To receive incoming remote calls for this interface, the client MUST implement this interface (09829352-87c2-418d-8d79-4133969a489d). It MUST then specify an object that implements this interface to the [IAppHostAdminManager::SetMetadata\(\)](#) method by using a bstrMetadataName of "changeHandler".

The server then calls this object when the administration system detects a change. The administration system is free to determine the supported time period during which the changes are detected and conveyed through this interface. The time period is either the lifetime of the administration system or a shorter time period.

The IAppHostChangeHandler interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
OnSectionChanges	Opnum: 3

3.1.4.2.1 OnSectionChanges (Opnum 3)

The OnSectionChanges method is called by the server by using an RPC_REQUEST packet. This method is called when a change in the path of the administration system hierarchy is detected. The callee (the client-implemented object) can react to this notification as it determines. It can return any error and the server MUST ignore it.

```
HRESULT OnSectionChanges(  
    [in] BSTR bstrSectionName,  
    [in] BSTR bstrConfigPath  
);
```

bstrSectionName: The name of the [IAppHostElement](#) on the server that changed. A server is free to not implement this parameter and always passes NULL.

bstrConfigPath: The path in the hierarchy where the change was detected by the administration system.

Return Values: The return value MUST be ignored by the server.

3.1.4.3 IAppHostChildElementCollection

The IAppHostChildElementCollection **interface** provides methods that allow access to any fixed child [IAppHostElement](#) objects that are contained by the specific IAppHostElement object that provides this interface through [IAppHostElement::ChildElements\(\)](#).

An IAppHostElement can contain any number of fixed, uniquely named child IAppHostElement objects. This IAppHostChildElementCollection provides access to these child elements.

The IAppHostChildElementCollection interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4

3.1.4.3.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the number of child [IAppHostElement](#) objects in the specific [IAppHostChildElementCollection](#) object.

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcCount  
);
```

pcCount: Receives a count of the number of child elements that are contained in the specified IAppHostChildElementCollection object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.3.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the requested child [IAppHostElement](#) object that is contained in the specific [IAppHostChildElementCollection](#) object.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT cIndex,  
    [out, retval] IAppHostElement** ppElement  
);
```

cIndex: A client-specified VARIANT that specifies the specific child IAppHostElement that is to be returned in the method. If the VARIANT is of type integer, the value represents a zero-based index that specifies the index of the child element to return, where 0 indicates the first child, 1 the second, and so on. The VARIANT can also be of the BSTR type, and if so, represents a string name that specifies the exact name of the child element to return.

ppElement: Receives a pointer to the child IAppHostElement that matches the criteria that is set by the cIndex parameter.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppElement MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585 ERROR_INVALID_INDEX	The index specified by cIndex is invalid.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.4 IAppHostCollectionSchema

The IAppHostCollectionSchema provides methods that describe the schema and constraints that apply to a specific [IAppHostElementCollection](#) from which this object was retrieved.

The IAppHostCollectionSchema **interface** inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
AddElementNames	Opnum: 3
GetAddElementSchema	Opnum: 4
RemoveElementSchema	Opnum: 5
ClearElementSchema	Opnum: 6
IsMergeAppend	Opnum: 7
GetMetadata	Opnum: 8
DoesAllowDuplicates	Opnum: 9

3.1.4.4.1 AddElementNames (Opnum 3)

The AddElementNames method is received by the server in an RPC_REQUEST packet. In response, the server returns a comma-delimited list of names that are supported by the administration system as names to [IAppHostElement](#) objects that are collection items of an [IAppHostElementCollection](#) object. An administration system typically supports only one name for the IAppHostElement objects that are contained in the collection. However, it could support more names in certain conditions; in which case, all the names are returned by using this method.

```
[propget] HRESULT AddElementNames(
    [out, retval] BSTR* pbstrElementName
);
```

pbstrElementName: Returns the comma-delimited string that contains the names of all supported names of IAppHostElement objects that are contained in the IAppHostElementCollection from which the specified [IAppHostCollectionSchema](#) was retrieved.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrElementName MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.4.2 GetAddElementSchema (Opnum 4)

The GetAddElementSchema method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostElementSchema](#) that represents the schema and constraints of the [IAppHostElement](#). The IAppHostElement can be a collection item of the specified [IAppHostElementCollection](#) from which the specified [IAppHostCollectionSchema](#) was retrieved and whose name matches the specified name in the method call.

```
HRESULT GetAddElementSchema(  
    [in] BSTR bstrElementName,  
    [out, retval] IAppHostElementSchema** ppSchema  
);
```

bstrElementName: The name of the IAppHostElement that is contained in the specified collection. It is one of the names that is returned in the [AddElementNames](#) method.

ppSchema: Returns the IAppHostElementSchema schema object that is associated with the specified element name.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSchema MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585	The element specified by bstrElementName cannot be found.

Return value/code	Description
ERROR_INVALID_INDEX	

3.1.4.4.3 RemoveElementSchema (Opnum 5)

The RemoveElementSchema method is received by the server in an RPC_REQUEST packet. In response, the server returns the [IAppHostElementSchema](#) that represents the schema and constraints of an optionally supported "directive IAppHostElement". This directive element can be used by the administration system to control the behavior of the specific [IAppHostElementCollection](#) from which the specified [IAppHostCollectionSchema](#) was retrieved.

```
[propget] HRESULT RemoveElementSchema(  
    [out, retval] IAppHostElementSchema** ppSchema  
);
```

ppSchema: Returns an IAppHostElementSchema object that is associated with the optionally supported directive [IAppHostElement](#).

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#).

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.4.4 ClearElementSchema (Opnum 6)

The ClearElementSchema method is received by the server in an RPC_REQUEST packet. In response, the server returns the [IAppHostElementSchema](#) that represents the schema and constraints of an optionally supported "directive IAppHostElement". This directive element can be used by the administration system to control behavior of the specific [IAppHostElementCollection](#) from which the specified [IAppHostCollectionSchema](#) was retrieved.

```
[propget] HRESULT ClearElementSchema(  
    [out, retval] IAppHostElementSchema** ppSchema  
);
```

ppSchema: Returns an IAppHostElementSchema object that is associated with the optionally supported directive [IAppHostElement](#).

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.4.5 IsMergeAppend (Opnum 7)

The IsMergeAppend method is received by the server in an RPC_REQUEST packet. In response, the server returns a Boolean that represents whether the [IAppHostElementCollection](#), from which the specified [IAppHostCollectionSchema](#) was retrieved, will prepend collection [IAppHostElement](#) objects from lower (deeper) in the hierarchy of the administration system with IAppHostElement objects from higher (shallower) in the hierarchy of the administration system.

If the value is false, lower IAppHostElement objects are prepended; otherwise, they are appended.

```
[propget] HRESULT IsMergeAppend(
    [out, retval] VARIANT_BOOL* pfIsMergeAppend
);
```

pfIsMergeAppend: The Boolean value that represents the append or prepend behavior.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.4.6 GetMetadata (Opnum 8)

The GetMetadata method is currently reserved for future use. The server MUST return ERROR_NOT_SUPPORTED (as defined in [\[MS-ERREF\]](#)) to indicate that the method isn't implemented.

```
HRESULT GetMetadata(
    [in] BSTR bstrMetadataType,
    [out, retval] VARIANT* pValue
);
```

bstrMetadataType: The name of the metadata that is retrieved.

pValue: Contains the value of the metadata being retrieved.

Return Values: The server MUST return ERROR_NOT_SUPPORTED (as defined in [MS-ERREF]) to indicate that the method isn't implemented.

3.1.4.4.7 DoesAllowDuplicates (Opnum 9)

The DoesAllowDuplicates method is received by the server in an RPC_REQUEST packet. In response, the server returns whether (through a Boolean) the [IAppHostElementCollection](#) from which the specified [IAppHostCollectionSchema](#) was retrieved allows duplicate [IAppHostElement](#) objects to exist in the collection. The concept of "duplicate" is implementation-specific.

```
[propget] HRESULT DoesAllowDuplicates(  
    [out, retval] VARIANT BOOL* pfAllowDuplicates  
);
```

pfAllowDuplicates: The Boolean value that represents whether duplicates are supported.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [MS-ERREF]. The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.5 IAppHostConfigException

Methods that receive an error when they access the administration system can retrieve more specific information about why the error occurred by using the [IAppHostConfigException](#) **interface**.

The IAppHostConfigException interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
LineNumber	Opnum: 3
FileName	Opnum: 4
ConfigPath	Opnum: 5
ErrorLine	Opnum: 6
PreErrorLine	Opnum: 7
PostErrorLine	Opnum: 8
ErrorString	Opnum: 9

3.1.4.5.1 LineNumber (Opnum 3)

The LineNumber (opnum 3) method is received by the server in an RPC_REQUEST packet. In response, the server returns a specific line number that might provide more detail regarding the location of the error in the hierarchy of the administration system.

```
[propget] HRESULT LineNumber(  
    [out, retval] unsigned long* pcLineNumber  
);
```

pcLineNumber: Contains the line number.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.5.2 FileName (Opnum 4)

The FileName method is received by the server in an RPC_REQUEST packet. In response, the server returns a file name that can provide more detail regarding the location of the error in the hierarchy of the administration system.

```
[propget] HRESULT FileName(  
    [out, retval] BSTR* pbstrFileName  
);
```

pbstrFileName: Contains the file name.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrFileName MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.5.3 ConfigPath (Opnum 5)

The ConfigPath method is received by the server in an RPC_REQUEST packet. In response, the server returns a path in the supported hierarchy of the administration system that contains the error.

```
[propget] HRESULT ConfigPath(  
    [out, retval] BSTR* pbstrConfigPath  
);
```

pbstrConfigPath: Contains the hierarchy path.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrConfigPath MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.5.4 ErrorLine (Opnum 6)

The ErrorLine method is received by the server in an RPC_REQUEST packet. In response, the server returns a textual representation of the specific data in the administration system that is causing the error.

```
[propget] HRESULT ErrorLine(  
    [out, retval] BSTR* pbstrErrorLine  
);
```

pbstrErrorLine: Contains the error data.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrErrorLine MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057	One or more parameters are incorrect or null.

Return value/code	Description
ERROR_INVALID_PARAMETER	
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.5.5 PreErrorLine (Opnum 7)

The PreErrorLine method is received by the server in an RPC_REQUEST packet. In response, the server returns a textual representation of the data that precedes the specific data in the administration system that is causing the error.

```
[propget] HRESULT PreErrorLine(
    [out, retval] BSTR* pbstrPreErrorLine
);
```

pbstrPreErrorLine: Contains the error data.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrPreErrorLine MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.5.6 PostErrorLine (Opnum 8)

The PostErrorLine method is received by the server in an RPC_REQUEST packet. In response, the server returns a textual representation of the data that follows the specific data in the administration system that is causing the error.

```
[propget] HRESULT PostErrorLine(
    [out, retval] BSTR* pbstrPostErrorLine
);
```

pbstrPostErrorLine: Contains the error data.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrPostErrorLine MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table

describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.5.7 ErrorString (Opnum 9)

The ErrorString method is received by the server in an RPC_REQUEST packet. In response, the server returns a description of the error that occurred in the administration system.

```
[propget] HRESULT ErrorString(  
    [out, retval] BSTR* pbstrErrorString  
);
```

pbstrErrorString: Contains the error description.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrErrorString MUST NOT be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.6 IAppHostConfigFile

The IAppHostConfigFile interface provides methods for direct access to a container of [IAppHostElement](#) objects in the administration system. The administration system implements a path hierarchy that allows for IAppHostElement objects to be defined at multiple places. This path hierarchy is conceptually split into two levels. One is the level that is represented by IAppHostConfigFile.

Each IAppHostConfigFile maps to a specific path (for example, "MACHINE/WEBROOT"). Within each IAppHostConfigFile, individual IAppHostElement objects exist, which are the base objects of the administration system. Optionally, the IAppHostConfigFile can also support subpaths within it.

The IAppHostConfigFile **interface** inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
ConfigPath	Opnum: 3
FilePath	Opnum: 4
Locations	Opnum: 5
GetAdminSection	Opnum: 6
GetMetadata	Opnum: 7
SetMetadata	Opnum: 8
ClearInvalidSections	Opnum: 9
RootSectionGroup	Opnum: 10

3.1.4.6.1 ConfigPath (Opnum 3)

The ConfigPath method is received by the server in an RPC_REQUEST packet. In response, the server returns the hierarchy path that is represented by the specific [IAppHostConfigFile](#).

```
[propget] HRESULT ConfigPath(  
    [out, retval] BSTR* pbstrConfigPath  
);
```

pbstrConfigPath: Contains the hierarchy path.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrConfigPath is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.6.2 FilePath (Opnum 4)

The FilePath method is received by the server in an RPC_REQUEST packet. In response, the server returns the server operating system file path that corresponds to the specific [IAppHostConfigFile](#), if that path applies to the administration system implementation.

```
[propget] HRESULT FilePath(
    [out, retval] BSTR* pbstrFilePath
);
```

pbstrFilePath: Contains the file path of the IAppHostConfigFile.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrFilePath is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.6.3 Locations (Opnum 5)

The Locations method is received by the server in an RPC_REQUEST packet. In response, the server returns any hierarchy subpaths that exist in the specified [IAppHostConfigFile](#). These subpaths are returned in the form of a collection object that implements the [IAppHostConfigLocationCollection](#).

```
[propget] HRESULT Locations(
    [out, retval] IAppHostConfigLocationCollection** ppLocations
);
```

ppLocations: Contains a collection object that contains all the subpaths that are available in the specified IAppHostConfigFile.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppLocations is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070002	A server resource (for example, a file on a disk) could not be found.

Return value/code	Description
ERROR_FILE_NOT_FOUND	
0X80070005 ERROR_ACCESS_DENIED	Access to a server resource (for example, a file on a disk) was denied.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.6.4 GetAdminSection (Opnum 6)

The GetAdminSection method is received by the server in an RPC_REQUEST packet. In response, the server returns a single [IAppHostElement](#) at the specified hierarchy path in the specific [IAppHostConfigFile](#). The section is not a merge and is instead a single IAppHostElement.

If no IAppHostElement object exists at the specified path, the implementation MAY return either an error or an empty IAppHostElement object (the default).

This method is used to get a specific administration section.

```
HRESULT GetAdminSection(
    [in] BSTR bstrSectionName,
    [in] BSTR bstrPath,
    [out, retval] IAppHostElement** ppAdminSection
);
```

bstrSectionName: The name of the IAppHostElement object to retrieve.

bstrPath: The hierarchy path of the IAppHostElement object to retrieve. It MUST be a path that is supported in the specific IAppHostConfigFile for the method to succeed and return an IAppHostElement object.

ppAdminSection: Contains an IAppHostElement from the specific IAppHostConfigFile.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppAdminSection is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.
0X80070002 ERROR_FILE_NOT_FOUND	The server resource (for example, a file or database) corresponding to the path bstrPath could not be found.
0X80070005	Access to the server resource (for example, a file or database)

Return value/code	Description
ERROR_ACCESS_DENIED	corresponding to the path bstrPath was denied.
0X00000002 ERROR_PATH_NOT_FOUND	The section specified by bstrPath is not supported.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.6.5 GetMetadata (Opnum 7)

The GetMetadata method is currently reserved for future use. The server MUST return ERROR_NOT_SUPPORTED (as defined in [\[MS-ERREF\]](#)) to indicate that the method isn't implemented.

```
HRESULT GetMetaData(
    [in] BSTR bstrMetadataType,
    [out, retval] VARIANT* pValue
);
```

bstrMetadataType: The name of the metadata property to fetch.

pValue: Returns the value of the specified metadata property. The type depends on the property that is fetched.

Return Values: The server MUST return ERROR_NOT_SUPPORTED (as defined in [\[MS-ERREF\]](#)) to indicate that the method isn't implemented.

3.1.4.6.6 SetMetadata (Opnum 8)

The SetMetadata method is currently reserved for future use. The server MUST return ERROR_NOT_SUPPORTED (as defined in [\[MS-ERREF\]](#)) to indicate that the method isn't implemented.

```
HRESULT SetMetadata(
    [in] BSTR bstrMetadataType,
    [in] VARIANT value
);
```

bstrMetadataType: The name of the metadata property to set.

value: The value of the metadata property to set.

Return Values: The server MUST return ERROR_NOT_SUPPORTED (as defined in [\[MS-ERREF\]](#)) to indicate that the method isn't implemented.

3.1.4.6.7 ClearInvalidSections (Opnum 9)

The ClearInvalidSections method is received by the server in an RPC_REQUEST packet. In response, the server MAY clear any invalid [IAppHostElement](#) objects that exist in the specific [IAppHostConfigFile](#).

```
HRESULT ClearInvalidSections();
```

This method has no parameters.

Return Values: This method MUST return S_OK (0x00000000) on success, and a failure result (as specified in [\[MS-ERREF\]](#) section 2) on failure. All failure results MUST be treated identically. The following table specifies failure results specific to this method.

Return value/code	Description
0x80070032 ERROR_NOT_SUPPORTED	The request is not supported.

When processing this call, the server MUST do the following:

- If clearing invalid sections is not supported, the server MUST return ERROR_NOT_SUPPORTED.

3.1.4.6.8 RootSectionGroup (Opnum 10)

The RootSectionGroup method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostSectionGroup](#) object, which represents a declaration of [IAppHostElement](#) objects that apply to the specified [IAppHostConfigFile](#) and that potentially apply to other [IAppHostConfigFile](#) at deeper hierarchy paths than the current file.

A declaration means that an [IAppHostElement](#) of the name in the declaration MAY exist in the specified [IAppHostConfigFile](#) or potentially in deeper paths. A declaration is NOT a definition/instance. A declaration only controls whether an actual [IAppHostElement](#) instance is supported.

This function returns the section group object in the configSections section group that defines the root section group for the current section.

```
[propget] HRESULT RootSectionGroup(  
    [out, retval] IAppHostSectionGroup** ppSectionGroups  
);
```

ppSectionGroups: Contains an [IAppHostSectionGroup](#) that contains declarations.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSectionGroups is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.7 IAppHostConfigLocation

The [IAppHostConfigLocation](#) **interface** provides methods that access the [IAppHostElement](#) objects that are contained in a specific hierarchy subpath in a specified [IAppHostConfigFile](#). [IAppHostConfigFile](#) maps to a specific hierarchy path in the administration system, as specified in section 3.1.4.6.

Each IAppHostConfigFile can optionally contain subpaths within it. Each subpath is represented by an IAppHostConfigLocation object. The object contains a collection of IAppHostElement objects, with the guarantee that each IAppHostConfigLocation contains at most one IAppHostElement object of the same name (in other words, the IAppHostElement object name is a key into the collection).

The IAppHostConfigLocation interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Path	Opnum: 3
Count	Opnum: 4
Item	Opnum: 5
AddConfigSection	Opnum: 6
DeleteConfigSection	Opnum: 7

3.1.4.7.1 Path (Opnum 3)

The Path method is received by the server in an RPC_REQUEST packet. In response, the server returns the subpath for the specific [IAppHostConfigLocation](#).

```
[propget] HRESULT Path(  
    [out, retval] BSTR* pbstrLocationPath  
);
```

pbstrLocationPath: Contains the path for the specific IAppHostConfigLocation. If the path is "" (an empty string), it represents the same path as the containing [IAppHostConfigFile](#).

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrLocationPath is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.7.2 Count (Opnum 4)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the number of [IAppHostElement](#) objects that are defined or exist at the specified [IAppHostConfigLocation](#).

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcCount  
);
```

pcCount: The integer count of the number of IAppHostElement objects at the specified IAppHostConfigLocation.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.7.3 Item (Opnum 5)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the [IAppHostElement](#) object that corresponds to the specific index parameter.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT cIndex,  
    [out, retval] IAppHostElement** ppSection  
);
```

cIndex: A VARIANT, which is used to specify the IAppHostElement object to retrieve. If the VARIANT is of type integer, the index is a zero-based index to the collection of IAppHostElement objects, where 0 indicates the first IAppHostElement object, 1 the second, and so on. If the VARIANT is of type string, the index is the name of the IAppHostElement object that is retrieved.

ppSection: Contains an IAppHostElement object, if found.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSection is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by cIndex is invalid or the element with name specified by cIndex could not be found.

Return value/code	Description
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.7.4 AddConfigSection (Opnum 6)

The AddConfigSection method is received by the server in an RPC_REQUEST packet. In response, the server attempts to create a new empty [IAppHostElement](#) and add it to the specified [IAppHostConfigLocation](#). The server MAY choose to create the IAppHostElement object in memory only and not persist it to permanent storage, such as a disk file, until later.

```
HRESULT AddConfigSection(
    [in] BSTR bstrSectionName,
    [out, retval] IAppHostElement** ppAdminElement
);
```

bstrSectionName: The name of the new IAppHostElement section to add.

ppAdminElement: Contains a newly created IAppHostElement.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppAdminElement is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X800700B7 ERROR_ALREADY_EXISTS	A configuration element with the name specified by bstrSectionName already exists.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.

3.1.4.7.5 DeleteConfigSection (Opnum 7)

The DeleteConfigSection method is received by the server in an RPC_REQUEST packet. In response, the server attempts to delete the [IAppHostElement](#) of the specified index.

```
HRESULT DeleteConfigSection(
```

```
[in] VARIANT cIndex
);
```

cIndex: A VARIANT, which is used to specify the IAppHostElement object to delete. If the VARIANT is of type integer, the index is a zero-based index to the collection of IAppHostElement objects, where 0 indicates the first IAppHostElement object, 1 the second, and so on. If the VARIANT is of type string, the index is the name of the IAppHostElement object being retrieved.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by cIndex is invalid, or the element with name specified by cIndex could not be found.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.8 IAppHostConfigLocationCollection

The IAppHostConfigLocationCollection **interface** provides methods that access the collection of subpaths that are available in the specified [IAppHostConfigFile](#).

The IAppHostConfigLocationCollection interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4
AddLocation	Opnum: 5
DeleteLocation	Opnum: 6

3.1.4.8.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the count of subpaths (each represented by the [IAppHostConfigLocation](#)) in the specified collection.

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcCount  
);
```

pcCount: Contains a count of the number of subpaths in the specified collection.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.8.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the specific [IAppHostConfigLocation](#) that represents the subpath being retrieved.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT varIndex,  
    [out, retval] IAppHostConfigLocation** ppLocation  
);
```

varIndex: A VARIANT, which is used to specify the IAppHostConfigLocation object to retrieve. If the VARIANT is of type integer, the index is a zero-based index to the collection of [IAppHostElement](#) objects, where 0 indicates the first IAppHostElement object, 1 the second, and so on. If the VARIANT is of type string, the index is the name of the subpath being retrieved.

ppLocation: Contains a subpath that represents IAppHostConfigLocation.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppLocation is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

Return value/code	Description
0X80070585 ERROR_INVALID_INDEX	The integer index specified by cIndex is invalid.
0X00000002 ERROR_PATH_NOT_FOUND	The subpath with name specified by cIndex could not be found.

3.1.4.8.3 AddLocation (Opnum 5)

The AddLocation method is received by the server in an RPC_REQUEST packet. In response, the server attempts to create a new subpath container in the [IAppHostConfigFile](#) that provides the specified [IAppHostConfigLocationCollection](#).

```
HRESULT AddLocation(
    [in] BSTR bstrLocationPath,
    [out, retval] IAppHostConfigLocation** ppNewLocation
);
```

bstrLocationPath: The new subpath to add.

ppNewLocation: Contains a new subpath container [IAppHostConfigLocation](#) object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppNewLocation is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070005 ERROR_ACCESS_DENIED	The instance is not editable.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X800700B7 ERROR_ALREADY_EXISTS	The location path specified by bstrLocationPath cannot be added since it already exists.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X800700A1 ERROR_BAD_PATHNAME	The server resource (for example, a file or database) corresponding to the path bstrLocationPath could not be found.

3.1.4.8.4 DeleteLocation (Opnum 6)

The DeleteLocation method is received by the server in an RPC_REQUEST packet. In response, the server attempts to delete the specific subpath container (the [IAppHostConfigLocation](#) object).

```
HRESULT DeleteLocation(  
    [in] VARIANT cIndex  
);
```

cIndex: A VARIANT, which is used to specify the IAppHostConfigLocation object to delete. If the VARIANT is of type integer, the index is a zero-based index to the collection of [IAppHostElement](#) objects, where 0 indicates the first IAppHostElement object, 1 the second, and so on. If the VARIANT is of type string, the index is the name of the subpath being deleted.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070005 ERROR_ACCESS_DENIED	The instance is not editable.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by cIndex is invalid, or the location with name specified by cIndex could not be found.

3.1.4.9 IAppHostConfigManager

The IAppHostConfigManager **interface** provides methods that allow access to the available hierarchy paths and the [IAppHostElement](#) objects that are defined within.

The IAppHostConfigManager interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
GetConfigFile	Opnum: 3
GetUniqueConfigPath	Opnum: 4

3.1.4.9.1 GetConfigFile (Opnum 3)

The GetConfigFile method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostConfigFile](#) for the specific hierarchy path.

The administration system implementation can choose to fail if the specified hierarchy path does not have an IAppHostConfigFile container for it. Or it can choose to succeed and create an empty IAppHostConfigFile container instead.

```
HRESULT GetConfigFile(  
    [in] BSTR bstrConfigPath,  
    [out, retval] IAppHostConfigFile** ppConfigFile
```

);

bstrConfigPath: The hierarchy path for the IAppHostConfigFile to retrieve.

ppConfigFile: Contains an IAppHostConfigFile object for the specified path.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppConfigFile is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.
0X80070002 ERROR_FILE_NOT_FOUND	The server resource (for example, a file or database) corresponding to bstrConfigPath could not be found.
0X80070005 ERROR_ACCESS_DENIED	Access to a server resource (for example, a file on a disk) was denied.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.9.2 GetUniqueConfigPath (Opnum 4)

The GetUniqueConfigPath method is received by the server in an RPC_REQUEST packet. In response, the server returns the deepest hierarchy path (up to the specified hierarchy path) that contains a unique set of [IAppHostElement](#) objects. For example:

Assume: At hierarchy path A, a set of IAppHostElement objects exist.

Assume: At hierarchy path B (deeper than A), the identical set of objects exists.

Given these assumptions, GetUniqueConfigPath(B) returns path A. In other words, the method returns the shallowest path that contains the identical set of IAppHostElement objects as the specified path.

```
HRESULT GetUniqueConfigPath(  
    [in] BSTR bstrConfigPath,  
    [out, retval] BSTR* pbstrUniquePath  
);
```

bstrConfigPath: The hierarchy path for which to find the shallowest equivalent path.

pbstrUniquePath: Contains the shallowest equivalent path.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrUniquePath is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.
0X00000002 ERROR_PATH_NOT_FOUND	A server resource (for example, a file on a disk) could not be found.
0X80070005 ERROR_ACCESS_DENIED	Access to a server resource (for example, a file on a disk) was denied.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.10 IAppHostConstantValue

The IAppHostConstantValue **interface** provides methods that access the string names of a specific constant and its corresponding value.

The IAppHostConstantValue interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Name	Opnum: 3
Value	Opnum: 4

3.1.4.10.1 Name (Opnum 3)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server provides the string name of the specified constant.

```
[propget] HRESULT Name(  
    [out, retval] BSTR* pbstrName  
);
```

pbstrName: Contains the name of the constant value.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.10.2 Value (Opnum 4)

The Value method is received by the server in an RPC_REQUEST packet. In response, the server returns the integer value of the constant.

```
[propget] HRESULT Value(
    [out, retval] DWORD* pdwValue
);
```

pdwValue: Contains the integer value of the specified constant.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.11 IAppHostConstantValueCollection

The IAppHostConstantValueCollection **interface** provides methods that access a collection of constant values.

The IAppHostConstantValueCollection interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4

3.1.4.11.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the count of constant values in the specified [IAppHostConstantValueCollection](#).

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcCount  
);
```

pcCount: Contains the count of constant values.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.11.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the specified constant value.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT cIndex,  
    [out, retval] IAppHostConstantValue** ppConstantValue  
);
```

cIndex: A VARIANT that represents the constant value to retrieve. If the VARIANT is of type integer, the index is a zero-based index to the collection of [IAppHostElement](#) objects, where 0 indicates the first IAppHostElement object, 1 the second, and so on. If the VARIANT is of type string, it is the name of the constant value.

ppConstantValue: Contains the constant value.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppConstantValue is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.

Return value/code	Description
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by cIndex is invalid, or the element with name specified by cIndex could not be found.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.12 IAppHostElement

The IAppHostElement **interface** provides methods to the base administration system object of this whole project.

The administration system is a container of IAppHostElement objects. IAppHostElement objects are primarily an *n*-way-tree object, with each IAppHostElement object primarily containing:

- Zero or more [IAppHostProperty](#) objects. Each has its own unique fixed name.
- Zero or more child IAppHostElement objects (hence the *n*-way-tree description). Each has its own unique fixed name.
- An optional collection of zero or more special child IAppHostElement objects that are called collection objects. Each typically has the same fixed name.
- Zero or more [IAppHostMethod](#) objects, which are additional extension methods that can be executed on the specified IAppHostElement object.

The IAppHostElement interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Name	Opnum: 3
Collection	Opnum: 4
Properties	Opnum: 5
ChildElements	Opnum: 6
GetMetadata	Opnum: 7
SetMetadata	Opnum: 8
Schema	Opnum: 9
GetElementByName	Opnum: 10
GetPropertyByName	Opnum: 11
Clear	Opnum: 12
Methods	Opnum: 13

3.1.4.12.1 Name (Opnum 3)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server returns the name of the [IAppHostElement](#) object.

```
[propget] HRESULT Name(  
    [out, retval] BSTR* pbstrName  
);
```

pbstrName: Contains the name of the IAppHostElement.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.12.2 Collection (Opnum 4)

The Collection method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostElementCollection](#) that represents a collection of "collection IAppHostElement" objects. If the specific [IAppHostElement](#) does not support this type of child object, it indicates this in the return.

```
[propget] HRESULT Collection(  
    [out, retval] IAppHostElementCollection** ppCollection  
);
```

ppCollection: Contains an IAppHostElementCollection that represents the collection child elements. If the specific IAppHostElement does not support this type of child element, this parameter is NULL.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

Return value/code	Description
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.12.3 Properties (Opnum 5)

The Properties method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostPropertyCollection](#) that contains the [IAppHostProperty](#) objects that are available for this [IAppHostElement](#).

```
[propget] HRESULT Properties(
    [out, retval] IAppHostPropertyCollection** ppProperties
);
```

ppProperties: Contains the IAppHostPropertyCollection that represents the collection of IAppHostProperty objects that are available.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppProperties is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.12.4 ChildElements (Opnum 6)

The ChildElements method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostChildElementCollection](#) that contains child [IAppHostElement](#) objects if any child IAppHostElement objects exist.

```
[propget] HRESULT ChildElements(
    [out, retval] IAppHostChildElementCollection** ppElements
);
```

ppElements: Contains an IAppHostChildElementCollection if there are child IAppHostElement objects in the specified IAppHostElement object. Otherwise, it is NULL.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.12.5 GetMetadata (Opnum 7)

The GetMetadata method is received by the server in an RPC_REQUEST packet. In response, the server returns the specific piece of named metadata.

```
HRESULT GetMetadata(
    [in] BSTR bstrMetadataType,
    [out, retval] VARIANT* pValue
);
```

bstrMetadataType: The name of the metadata property to fetch. Valid names are as follows.

Value	Meaning
"overrideMode"	A string that represents the explicit override mode of the specified IAppHostElement object. Override mode defines the locking behavior of an IAppHostElement object as it relates to other IAppHostElement objects at deeper hierarchy paths. The administration system can implement override rules, the selection of which is reflected by this metadata. string
"effectiveOverrideMod"	A string that represents the effective override mode of the specified IAppHostElement object. Override mode defines the locking behavior of an IAppHostElement object as it relates to other IAppHostElement objects at deeper hierarchy paths. The administration system can implement override rules, the selection of which is reflected by this metadata. string
"allowCommit"	A Boolean that represents whether the administration system supports changing or editing of the specified IAppHostElement object. bool
"deepestPathSet"	A string that represents the deepest hierarchy path from which the specified IAppHostElement object was derived. This is only meaningful if this IAppHostElement object is a merged object from multiple sources. This metadata is the path of the deepest source that was merged. string

Value	Meaning
"deepestFileNameSet"	A string that represents an operating system file path (if applicable) that corresponds to the deepest hierarchy path metadata "deepestPathSet". string
"deepestFileLineNumberSet"	An integer that represents a location within the corresponding metadata "deepestFileNameSet". uint32
"configSource"	A string that represents an optional redirection path from which the specified IAppHostElement object was retrieved. The path is specific to the administration system and is only applicable if the administration system supports alternate redirection paths. string
"childSource"	A string that represents an optional redirection path from which the specified IAppHostElement object was retrieved. The path is specific to the administration system and only applicable if the administration system supports alternate redirection paths. The difference between "configSource" and "childSource" is an implementation detail of the redirection support of the administration system. string
"lockItem"	A Boolean that represents whether the specified IAppHostElement object can be set at deeper hierarchy paths. If true, the item cannot be set deeper. bool
"lockAllElementsExcept"	A string that represents a comma-delimited list of child element names that are allowed to be set at deeper hierarchy paths. All other unlisted child elements are not allowed to be set at deeper paths. string
"lockElements"	A string that represents a comma-delimited list of child element names that are not allowed to be set at deeper hierarchy paths. string
"lockAllAttributesExcept"	A string that represents a comma-delimited list of property names that are allowed to be set at deeper hierarchy paths. All other unlisted properties are not allowed to be set at deeper paths. string
"lockAttributes"	A string that represents a comma-delimited list of property names that are not allowed to be set at deeper hierarchy paths. string

If SetMetadata() is called with another name, that metadata is stored and can be retrieved with GetMetadata().

pValue: Contains a VARIANT that represents the metadata that is retrieved.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.

Return value/code	Description
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property specified by bstrMetadataType is not supported.
0X00000008 E_OUTOFMEMORY	Not enough memory is available to process this command.

3.1.4.12.6 SetMetadata (Opnum 8)

The SetMetadata method is received by the server in an RPC_REQUEST packet. In response, the server returns the value of the metadata from the server administration system.

This method is used to set a metadata property.

```
HRESULT SetMetadata(
    [in] BSTR bstrMetadataType,
    [in] VARIANT value
);
```

bstrMetadataType: The name of the metadata property to set. Valid names are as follows.

Value	Meaning
"overrideMode"	A string that represents the explicit override mode of the specified IAppHostElement object. Override mode defines the locking behavior of an IAppHostElement object as it relates to other IAppHostElement objects at deeper hierarchy paths. The administration system is free to implement override rules, the selection of which is reflected by this metadata. string
"configSource"	A string that represents an optional redirection path from which the specified IAppHostElement object was retrieved. The path is specific to the administration system and only applicable if the administration system supports alternate redirection paths. string
"childSource"	A string that represents an optional redirection path from which the specified IAppHostElement object was retrieved. The path is specific to the administration system and only applicable if the administration system supports alternate redirection paths. The difference between "configSource" and "childSource" is totally an implementation detail of the redirection support of the administration system. string
"lockItem"	A Boolean that represents whether the specified IAppHostElement object can be set at deeper hierarchy paths. If true, the item cannot be set deeper. bool
"lockAllElementsExcept"	A string that represents a comma-delimited list of child element names that are allowed to be set at deeper hierarchy paths. All other unlisted child elements are not allowed to be set at deeper paths. string

Value	Meaning
"lockElements"	A string that represents a comma-delimited list of child element names that are not allowed to be set at deeper hierarchy paths. string
"lockAllAttributesExcept"	A string that represents a comma-delimited list of property names that are allowed to be set at deeper hierarchy paths. All other unlisted properties are not allowed to be set at deeper paths. string
"lockAttributes"	A string that represents a comma-delimited list of property names that are not allowed to be set at deeper hierarchy paths. string

If SetMetadata() is called with another name, that metadata is stored and can be retrieved with GetMetadata().

value: The value of the metadata property to set.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property specified by bstrMetadataType is not supported.
0X80070013 ERROR_INVALID_DATA	The data is invalid. The value that was attempted to be set is invalid against the schema.

3.1.4.12.7 Schema (Opnum 9)

The Schema method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostElementSchema](#) that represents the schema and constraints for the specified [IAppHostElement](#) object.

```
[propget] HRESULT Schema(
    [out, retval] IAppHostElementSchema** ppSchema
);
```

ppSchema: Contains schema of the specified IAppHostElement object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSchema is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error

conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.12.8 GetElementByName (Opnum 10)

The GetElementByName method is received by the server in an RPC_REQUEST packet. In response, the server returns the child [IAppHostElement](#) object with the specified name.

```
HRESULT GetElementByName(
    [in] BSTR bstrSubName,
    [out, retval] IAppHostElement** ppElement
);
```

bstrSubName: The name of the child element to retrieve.

ppElement: Contains the child IAppHostElement object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppElement is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585 ERROR_INVALID_INDEX	The child element specified by bstrSubName could not be found.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.12.9 GetPropertyByName (Opnum 11)

The GetPropertyByName method is received by the server in an RPC_REQUEST packet. In response, the server returns the [IAppHostProperty](#) of the specified name.

```
HRESULT GetPropertyByName(
    [in] BSTR bstrSubName,
    [out, retval] IAppHostProperty** ppProperty
);
```

);

bstrSubName: The name of the IAppHostProperty to fetch.

ppProperty: Contains an IAppHostProperty of the specified name.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppProperty is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585 ERROR_INVALID_INDEX	The property specified by bstrSubName could not be found.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.12.10 Clear (Opnum 12)

The Clear method is received by the server in an RPC_REQUEST packet. In response, the server clears the contents of the specified [IAppHostElement](#) object. The administration system can choose to either perform the clearing of memory or persist it directly to permanent storage.

```
HRESULT Clear();
```

This method has no parameters.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070021 ERROR_LOCK_VIOLATION	The element is not editable.

3.1.4.12.11 Methods (Opnum 13)

The Methods method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostMethodCollection](#), which is the collection of methods that are supported for the specific [IAppHostElement](#) object.

```
[propget] HRESULT Methods(  
    [out, retval] IAppHostMethodCollection** ppMethods  
);
```

ppMethods: Contains an IAppHostMethodCollection, which is a collection of methods that are supported for the specific IAppHostElement object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppMethods is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.13 IAppHostElementCollection

The IAppHostElementCollection interface provides methods that access a collection of "collection IAppHostElements".

"Collection IAppHostElements" are a special class of child [IAppHostElement](#) objects where all objects typically share the same name but contain different [IAppHostProperty](#) objects. This IAppHostElementCollection is a collection of these special objects.

The IAppHostElementCollection **interface** inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Returns the count of elements. Opnum: 3
Item	Returns the value that is associated with the element at the specified index. Opnum: 4
AddElement	Adds an element to the collection. Opnum: 5
DeleteElement	Deletes an element from the collection. Opnum: 6

Method	Description
Clear	Clears an element from the collection. Opnum: 7
CreateNewElement	Creates a new element in the collection. Opnum: 8
Schema	Returns the Document Object Model (DOM) description for the collection. Opnum: 9

3.1.4.13.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns a count of the number of [IAppHostElement](#) objects in the collection.

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcElementCount  
);
```

pcElementCount: Contains the number of elements that are contained in the element collection.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.13.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostElement](#) for the specified index.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT cIndex,  
    [out, retval] IAppHostElement** ppElement  
);
```

cIndex: A VARIANT that specifies the IAppHostElement object to return from the collection. If the VARIANT is of type integer, the index is a zero-based index to the collection, where 0 indicates the first IAppHostElement, 1 the second, and so on. If the VARIANT is of type IAppHostElement, the index is a "selector" IAppHostElement to the specified collection.

ppElement: Contains the collection IAppHostElement.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppElement is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by cIndex is invalid, or the IAppHostElement specified by cIndex doesn't match any element in the collection.

3.1.4.13.3 AddElement (Opnum 5)

The AddElement method is received by the server in an RPC_REQUEST packet. In response, the server attempts to add a newly created [IAppHostElement](#) object to the specified collection.

This function adds an element to the current collection of elements.

```
HRESULT AddElement(
    [in] IAppHostElement* pElement,
    [in, defaultvalue(-1)] int cPosition
);
```

pElement: The element to add to the collection.

cPosition: The position at which the element was added to the collection. If -1, the new element is appended. Otherwise, the position represents the zero-based index to the collection.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.
0X80070585 ERROR_INVALID_INDEX	The index specified by cPosition is invalid.

3.1.4.13.4 DeleteElement (Opnum 6)

The DeleteElement method is received by the server in an RPC_REQUEST packet. In response, the server deletes the [IAppHostElement](#) at the specified index.

```
HRESULT DeleteElement(  
    [in] VARIANT cIndex  
);
```

cIndex: A VARIANT that specifies the IAppHostElement object to return from the collection. If the VARIANT is of type integer, the index is a zero-based index to the collection, where 0 indicates the first IAppHostElement, 1 the second, and so on. If the VARIANT is of type IAppHostElement, the index is a "selector" IAppHostElement to the specified collection.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.
0X80070585 ERROR_INVALID_INDEX	The index specified by cIndex is invalid.

3.1.4.13.5 Clear (Opnum 7)

The Clear method is received by the server in an RPC_REQUEST packet. In response, the server clears the specified [IAppHostElementCollection](#) of all the collection [IAppHostElement](#) objects in it.

```
HRESULT Clear();
```

This method has no parameters.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.

3.1.4.13.6 CreateNewElement (Opnum 8)

The CreateNewElement method is received by the server in an RPC_REQUEST packet. In response, the server creates a new unattached [IAppHostElement](#) object that has the specified name. The name MUST be a supported name as defined by the [IAppHostCollectionSchema](#) of the specified [IAppHostElementCollection](#).

```
HRESULT CreateNewElement(
    [in, defaultvalue("")] BSTR bstrElementName,
    [out, retval] IAppHostElement** ppElement
);
```

bstrElementName: The name of the IAppHostElement to be created.

ppElement: Contains a new IAppHostElement object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppElement is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.
0X80070585 ERROR_INVALID_INDEX	The schema does not permit the creation of an element with name bstrElementName.

3.1.4.13.7 Schema (Opnum 9)

The Schema method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostCollectionSchema](#) for the specified [IAppHostElementCollection](#), which represents the schema and constraints of the specified collection.

```
[propget] HRESULT Schema(  
    [out, retval] IAppHostCollectionSchema** ppSchema  
);
```

ppSchema: Contains the schema object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSchema is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.14 IAppHostElementSchema

The IAppHostElementSchema **interface** provides methods that access the schema and constraints of a specific [IAppHostElement](#) object.

The IAppHostElementSchema interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Name	Returns the name of the element entry. Opnum: 3
DoesAllowUnschemasizedProperties	Determines if the section allows unrecognized attributes. Opnum: 4
GetMetadata	Used to get a metadata property. Opnum: 5
CollectionSchema	Opnum: 6
ChildElementSchemas	Opnum: 7
PropertySchemas	Opnum: 8
IsCollectionDefault	Opnum: 9

3.1.4.14.1 Name (Opnum 3)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server returns the name of the corresponding [IAppHostElement](#) from which the [IAppHostElementSchema](#) was retrieved.

```
[propget] HRESULT Name(  
    [out, retval] BSTR* pbstrName  
);
```

pbstrName: Contains the name of the element.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.14.2 DoesAllowUnschematizedProperties (Opnum 4)

The DoesAllowUnschematizedProperties method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the corresponding [IAppHostElement](#) supports [IAppHostProperty](#) objects.

```
[propget] HRESULT DoesAllowUnschematizedProperties(  
    [out, retval] VARIANT_BOOL* pfAllowUnschematized  
);
```

pfAllowUnschematized: Contains a Boolean that indicates whether the element supports unschematized IAppHostProperty objects.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.14.3 GetMetadata (Opnum 5)

The GetMetadata method is received by the server in an RPC_REQUEST packet. In response, the server returns the specific element of named metadata.

```
HRESULT GetMetadata(  
    [in] BSTR bstrMetadataType,  
    [out, retval] VARIANT* pValue  
);
```

bstrMetadataType: The name of the metadata property to fetch. Valid names are as follows.

Value	Meaning
"extension"	Returns a string that identifies the server-side extension that implements this IAppHostElementSchema instance, if one exists, or NULL otherwise. The client can use this metadata property to query if the IAppHostElementSchema has a server-side extension. The actual contents of the string are implementation-specific and MUST be ignored by the client.

pValue: Contains a VARIANT that represents the metadata that is retrieved.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property specified by bstrMetadataType is not supported.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.14.4 CollectionSchema (Opnum 6)

The CollectionSchema method is received by the server in an RPC_REQUEST packet. If the specified [IAppHostElement](#) object supports child collection elements, the server returns the schema and constraints of the collection that is contained in the corresponding IAppHostElement object.

```
[propget] HRESULT CollectionSchema(  
    [out, retval] IAppHostCollectionSchema** ppCollectionSchema  
);
```

ppCollectionSchema: Contains an [IAppHostCollectionSchema](#).

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppCollectionSchema is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the

error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.14.5 ChildElementSchemas (Opnum 7)

The ChildElementSchemas method is received by the server in an RPC_REQUEST packet. In response, the server returns the schema and constraints of any child elements that are contained in the corresponding [IAppHostElement](#) object.

```
[propget] HRESULT ChildElementSchemas(  
    [out, retval] IAppHostElementSchemaCollection** ppChildSchemas  
);
```

ppChildSchemas: Contains the schema collection of the supported child IAppHostElement objects.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppChildSchemas is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.14.6 PropertySchemas (Opnum 8)

The PropertySchemas method is received by the server in an RPC_REQUEST packet. In response, the server returns the schema and constraints for the [IAppHostProperty](#) objects that are contained in the corresponding [IAppHostElement](#).

```
[propget] HRESULT PropertySchemas(  
    [out, retval] IAppHostPropertySchemaCollection** ppPropertySchemas  
);
```

ppPropertySchemas: Contains the collection of IAppHostProperty schema.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppPropertySchemas is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.14.7 IsCollectionDefault (Opnum 9)

The IsCollectionDefault method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the corresponding [IAppHostElement](#) object is also considered to be a supported default for other IAppHostElement objects in the administration system.

```
[propget] HRESULT IsCollectionDefault(
    [out, retval] VARIANT_BOOL* pfIsCollectionDefault
);
```

pfIsCollectionDefault: Contains a Boolean that indicates whether the corresponding IAppHostElement object is a default for other IAppHostElement objects.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.15 IAppHostElementSchemaCollection

The IAppHostElementSchemaCollection **interface** provides methods that access a collection of schema and constraints for child [IAppHostElement](#) objects that are supported by the corresponding IAppHostElement object.

The IAppHostElementSchemaCollection interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4

3.1.4.15.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the count of [IAppHostElementSchema](#) objects that are contained in the collection.

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcCount  
);
```

pcCount: Contains the count in the collection.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.15.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the [IAppHostElementSchema](#) that is specified by the specific index.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT cIndex,  
    [out, retval] IAppHostElementSchema** ppElementSchema  
);
```

cIndex: A VARIANT that specifies the IAppHostElementSchema to retrieve from the collection. If the VARIANT is of type integer, this is a zero-based index to the collection. If the VARIANT is of type string, this is a string index that represents the name of the IAppHostElementSchema that is being retrieved.

ppElementSchema: Contains the IAppHostElementSchema that is being selected.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppElementSchema is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by cIndex is invalid, or there is no IAppHostElementSchema instance in the collection with a name as specified by cIndex.

3.1.4.16 IAppHostMappingExtension

The IAppHostMappingExtension **interface** provides methods that access the path hierarchy mapping system of the administration system.

The administration system implements a path hierarchy system that maps paths to potential [IAppHostElement](#) containers. Some details of the path hierarchy are exposed to the user in this IAppHostMappingExtension.

The IAppHostMappingExtension interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
GetSiteNameFromSiteId	Opnum: 3
GetSiteIdFromSiteName	Opnum: 4
GetSiteElementFromSiteId	Opnum: 5
MapPath	Opnum: 6

3.1.4.16.1 GetSiteNameFromSiteID (Opnum 3)

The GetSiteNameFromSiteId method is received by the server in an RPC_REQUEST packet. In response, the server returns a string name for the specific integer site ID, which is a concept that is implemented on the administration system (it is an implementation detail).

```
HRESULT GetSiteNameFromSiteId(  
    [in] DWORD dwSiteId,  
    [out, retval] BSTR* pbstrSiteName  
);
```

dwSiteId: The unique site ID number that is represented in a double-word format. Note that "0" is not a valid site ID.

pbstrSiteName: Contains the string that represents the unique site name for the specified site ID.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrSiteName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error

conditions that **MUST** be handled and the corresponding error codes. A server **MAY** return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070002 ERROR_FILE_NOT_FOUND	The configuration has no site with ID dwSiteId.
0X80070490 ERROR_NOT_FOUND	The configuration contains no sites.

3.1.4.16.2 GetSiteIdFromSiteName (Opnum 4)

The GetSiteIdFromSiteName method is received by the server in an RPC_REQUEST packet. In response, the server returns a unique integer ID for the specific site name. Site name and ID are implementation details of the administration system.

```
HRESULT GetSiteIdFromSiteName(
    [in] BSTR bstrSiteName,
    [out, retval] DWORD* pdwSiteId
);
```

bstrSiteName: The unique site name string.

pdwSiteId: Contains the double-word value that represents the unique site ID for the specified name. Note that "0" is NOT a valid site ID.

Return Values: The server **MUST** return zero if it successfully processes the message that is received from the client. If processing fails, the server **MUST** return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that **MUST** be handled and the corresponding error codes. A server **MAY** return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070002 ERROR_FILE_NOT_FOUND	The configuration has no site with name bstrSiteName.

Return value/code	Description
0X80070490 ERROR_NOT_FOUND	The configuration contains no sites.

3.1.4.16.3 GetSiteElementFromSiteID (Opnum 5)

The GetSiteElementFromSiteId method is received by the server in an RPC_REQUEST packet. In response, the server obtains the site section element from a specific site ID in order to access site configuration and properties.

```
HRESULT GetSiteElementFromSiteId(
    [in] DWORD dwSiteId,
    [out, retval] IAppHostElement** ppSiteElement
);
```

dwSiteId: The unique site ID number that is represented in a double-word format.

ppSiteElement: Returns an administration element object that represents the site element of the site section and includes properties, metadata, and methods.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070002 ERROR_FILE_NOT_FOUND	The given site ID could not be found.

3.1.4.16.4 MapPath (Opnum 6)

The MapPath method is received by the server in an RPC_REQUEST packet. In response, the server returns how the administration system maps the specific site name and virtual path into an optionally returned implementation-specific operating system physical path, and the server optionally returns the [IAppHostElement](#) objects that contain the definition of the mapping hierarchy that applies.

The "mapping hierarchy that applies" means describing the matching rules that the administration system uses to map a specific hierarchy path. Because there are two returned IAppHostElement objects, this allows the server protocol implementation to provide a two-level mapping system where a path is first matched to an application and then mapped to a virtual directory. Regardless of what these two concepts (application and virtual directory) represent, their mapping details are contained in the optionally returned parameters.

```

HRESULT MapPath(
    [in] BSTR bstrSiteName,
    [in] BSTR bstrVirtualPath,
    [out] BSTR* pbstrPhysicalPath,
    [out] IAppHostElement** ppVirtualDirectoryElement,
    [out] IAppHostElement** ppApplicationElement
);

```

bstrSiteName: The unique site name string.

bstrVirtualPath: A hierarchy path to map.

pbstrPhysicalPath: Returns an implementation-specific physical path in the file system to where the virtual path maps. Optional.

ppVirtualDirectoryElement: Returns the second-level mapped virtual directory that is matched in the mapping. Optional.

ppApplicationElement: Returns the first-level mapped application that is matched in the mapping. Optional.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. For each of the output parameters, in this case, they will not be NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.17 IAppHostMethod

The IAppHostMethod **interface** provides methods that access a custom method that is optionally supported on a specific [IAppHostElement](#) object.

An IAppHostElement object provides a means for an administration system to support custom-defined methods that can be executed against a specific IAppHostElement object. The methods are executed on the server side and the implementation of these custom methods is not exposed to the client.

The IAppHostMethod interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Name	Opnum: 3
Schema	Opnum: 4
CreateInstance	Opnum: 5

3.1.4.17.1 Name (Opnum 3)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server returns the name of the custom method.

```
[propget] HRESULT Name(  
    [out, retval] BSTR* pbstrName  
);
```

pbstrName: Contains the name of the method.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.17.2 Schema (Opnum 4)

The Schema method is received by the server in an RPC_REQUEST packet. In response, the server returns the schema and constraints for the specified custom [IAppHostMethod](#).

```
[propget] HRESULT Schema(  
    [out, retval] IAppHostMethodSchema** ppMethodSchema  
);
```

ppMethodSchema: Contains the schema of the specified IAppHostMethod.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppMethodSchema is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.17.3 CreateInstance (Opnum 5)

The CreateInstance method is received by the server in an RPC_REQUEST packet. In response, the server creates an instance object of the method that can be executed. This behavior is analogous to the stack frame of a native method call.

```
HRESULT CreateInstance(  
    [out, retval] IAppHostMethodInstance** ppMethodInstance  
);
```

ppMethodInstance: Contains the method instance.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppMethodInstance is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.18 IAppHostMethodCollection

The IAppHostMethodCollection **interface** provides methods that access a collection of supported [IAppHostMethod](#) objects.

The IAppHostMethodCollection interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4

3.1.4.18.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the count of [IAppHostMethod](#) objects in the collection.

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcCount  
);
```

pcCount: Contains the count of IAppHostMethod objects.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.18.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the [IAppHostMethod](#) that is specified by the index.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT cIndex,  
    [out, retval] IAppHostMethod** ppMethod  
);
```

cIndex: A VARIANT that specifies which IAppHostMethod is being selected from the collection. If the VARIANT is of type integer, the index is a zero-based index to the collection. If the VARIANT is of type string, the index is the name of the method being accessed.

ppMethod: Contains the IAppHostMethod that is being retrieved.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppMethod is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

Return value/code	Description
0X80070585 ERROR_INVALID_INDEX	The integer specified by cIndex is invalid, or there is no IappHostMethod instance in the collection with a name as specified by cIndex.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.19 IAppHostMethodInstance

The IAppHostMethodInstance interface provides methods that access a specific invocation instance of the corresponding [IAppHostMethod](#). The caller sets parameters and then executes the instance of the method.

The IAppHostMethodInstance **interface** inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Input	Opnum: 3
Output	Opnum: 4
Execute	Opnum: 5
GetMetadata	Opnum: 6
SetMetadata	Opnum: 7

3.1.4.19.1 Input (Opnum 3)

The Input method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostElement](#) in which input parameters can be specified for the specified method instance call. If the method does not support input parameters, no IAppHostElement is returned.

```
[propget] HRESULT Input(
    [out, retval] IAppHostElement** ppInputElement
);
```

ppInputElement: Contains the IAppHostElement that represents the input parameters.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057	One or more parameters are incorrect or null.

Return value/code	Description
ERROR_INVALID_PARAMETER	

3.1.4.19.2 Output (Opnum 4)

The Output method is received by the server in an RPC_REQUEST packet. In response, the server returns an [IAppHostElement](#) from which output parameters are retrieved after the specified method instance has been executed. If the method does not support output parameters, no IAppHostElement is returned.

```
[propget] HRESULT Output(
    [out, retval] IAppHostElement** ppOutputElement
);
```

ppOutputElement: Contains the IAppHostElement that represents the output parameters.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.19.3 Execute (Opnum 5)

The Execute method is received by the server in an RPC_REQUEST packet. In response, the server actually executes the specified custom method.

```
HRESULT Execute();
```

This method has no parameters.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.

3.1.4.19.4 GetMetaData (Opnum 6)

The GetMetadata method is received by the server in an RPC_REQUEST packet. In response, the server returns the specific piece of named metadata property. The metadata properties supported by this method are not mandated by this specification. The server MAY choose to support some metadata properties that are specific to its implementation. The server MAY also choose to not implement any properties.

```
HRESULT GetMetadata(  
    [in] BSTR bstrMetadataType,  
    [out, retval] VARIANT* pValue  
);
```

bstrMetadataType: The name of the implementation-specific metadata being retrieved.

pValue: A VARIANT that contains the metadata value.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 E_INVALIDARG	One or more parameters are incorrect or null.
0X00000008 E_OUTOFMEMORY	Not enough memory is available to process this command.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property specified by bstrMetadataType is not supported.

3.1.4.19.5 SetMetadata (Opnum 7)

The SetMetadata method is received by the server in an RPC_REQUEST packet. In response, the server sets the specific named metadata property. The metadata properties supported by this method are not mandated by this specification. The server MAY choose to support some metadata properties that are specific to its implementation. The server MAY also choose to not implement any properties.

```
HRESULT SetMetadata(  
    [in] BSTR bstrMetadataType,  
    [in] VARIANT value  
);
```

bstrMetadataType: The name of the implementation-specific metadata being set.

value: A VARIANT that contains the new value of metadata.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in

[\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 E_INVALIDARG	One or more parameters are incorrect or null.
0X00000008 E_OUTOFMEMORY	Not enough memory is available to process this command.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property specified by bstrMetadataType is not supported.

3.1.4.20 IAppHostMethodSchema

The IAppHostMethodSchema **interface** provides methods that access the schema and constraints of the corresponding [IAppHostMethod](#).

The IAppHostMethodSchema interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Name	Opnum: 3
InputSchema	Opnum: 4
OutputSchema	Opnum: 5
GetMetadata	Opnum: 6

3.1.4.20.1 Name (Opnum 3)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server returns the name of the specified [IAppHostMethod](#).

```
[propget] HRESULT Name(  
    [out, retval] BSTR* pbstrName  
);
```

pbstrName: Contains the name of the IAppHostMethod.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 E_OUTOFMEMORY	Not enough memory is available to process this command.

3.1.4.20.2 InputSchema (Opnum 4)

The InputSchema method is received by the server in an RPC_REQUEST packet. In response, the server returns the schema and constraints of the input parameters to the method call. This can be NULL if no input parameters are defined for the method.

```
[propget] HRESULT InputSchema(
    [out, retval] IAppHostElementSchema** ppInputSchema
);
```

ppInputSchema: Contains the input parameter schema.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.20.3 OutputSchema (Opnum 5)

The OutputSchema method is received by the server in an RPC_REQUEST packet. In response, the server returns the schema and constraints of the output parameters to the method call. This can be NULL if no output parameters are defined for the method.

```
[propget] HRESULT OutputSchema(
    [out, retval] IAppHostElementSchema** ppOutputSchema
);
```

ppOutputSchema: Contains the output parameter schema.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.20.4 GetMetadata (Opnum 6)

The GetMetadata method is received by the server in an RPC_REQUEST packet. In response, the server returns the named metadata for the specified method schema. The metadata properties supported by this method are not mandated by this specification. The server MAY choose to support some metadata properties that are specific to its implementation. The server MAY also choose to not implement any properties.

```
HRESULT GetMetadata(
    [in] BSTR bstrMetadataType,
    [out, retval] VARIANT* pValue
);
```

bstrMetadataType: The name of the metadata.

pValue: Contains the value of the metadata.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property requested by bstrMetadataType is not supported.

3.1.4.21 IAppHostPathMapper

The IAppHostPathMapper **interface** provides methods that are called by the server implementation when the server informs the client about hierarchy mapping decisions.

To receive incoming remote calls for this interface, the client MUST implement a **UUID** (e7927575-5cc3-403b-822e-328a6b904bee). It MUST then specify an object that implements this interface to the [IAppHostAdminManager::SetMetadata\(\)](#) method with a bstrMetadataName of "pathMapper".

As an administration system maps hierarchy paths to physical paths on the server, it optionally calls this client-supplied object that implements the IAppHostPathMapper interface. The implementer of this interface receives details of all mappings and can change the results of each mapping if required.

The IAppHostPathMapper interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
MapPath	Opnum: 3

3.1.4.21.1 MapPath (Opnum 3)

The MapPath method is called by the server in an RPC_REQUEST packet. In response, the client implementation receives the details of the specific mapping decision and optionally, can change the results by using its return.

```
HRESULT MapPath(  
    [in] BSTR bstrConfigPath,  
    [in] BSTR bstrMappedPhysicalPath,  
    [out, retval] BSTR* pbstrNewPhysicalPath  
);
```

bstrConfigPath: The hierarchy path being mapped.

bstrMappedPhysicalPath: The server side physical path that the administration system has determined maps to the specified hierarchy path.

pbstrNewPhysicalPath: Set to the new or updated physical path to use for the mapping. If the mapping stays the same, the client implementer returns the identical physical path that was passed in as *bstrMappedPhysicalPath*.

Return Values: The client MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.22 IAppHostProperty

The IAppHostProperty **interface** provides methods that access properties that can be contained under an [IAppHostElement](#) object.

IAppHostElement objects can contain zero or more IAppHostProperty objects. These IAppHostProperty objects typically represent a single setting or attribute set on an IAppHostElement but can be multiple settings or sets.

The IAppHostProperty interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Name	Opnum: 3
Value	"getter" Opnum: 4
Value	"setter" Opnum: 5
Clear	Opnum: 6
StringValue	Opnum: 7
Exception	Opnum: 8
GetMetadata	Opnum: 9
SetMetadata	Opnum: 10
Schema	Opnum: 11

3.1.4.22.1 Name (Opnum 3)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server returns the name of the specific [IAppHostProperty](#).

```
[propget] HRESULT Name(  
    [out, retval] BSTR* pbstrName  
);
```

pbstrName: Contains the name of the property.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.22.2 Value (Get) (Opnum 4)

The Value (Get) method is received by the server in an RPC_REQUEST packet. In response, the server returns the value of the specified property.

```
[propget] HRESULT Value(  
    [out, retval] VARIANT* pVariant  
);
```

pVariant: Contains the VARIANT value of the property.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.22.3 Value (Set) (Opnum 5)

The Value (Set) method is received by the server in an RPC_REQUEST packet. In response, the server sets the value of the specified property.

```
[propput] HRESULT Value(  
    [in] VARIANT value  
);
```

value: The new VARIANT value of the property.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.

Return value/code	Description
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.22.4 Clear (Opnum 6)

The Clear method is received by the server in an RPC_REQUEST packet. In response, the server clears the value of the specified property.

```
HRESULT Clear();
```

This method has no parameters.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.
0X80070585 ERROR_INVALID_INDEX	The schema does not define the property being cleared.

3.1.4.22.5 StringValue (Opnum 7)

The StringValue method is received by the server in an RPC_REQUEST packet. In response, the server returns a string representation of the value of the specified property.

```
[propget] HRESULT StringValue(
    [out, retval] BSTR* pbstrValue
);
```

pbstrValue: Contains the string value that represents the property value.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrValue is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.

Return value/code	Description
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070013 ERROR_INVALID_DATA	The property has a type that is not permitted by the schema.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.22.6 Exception (Opnum 8)

The Exception method is received by the server in an RPC_REQUEST packet. In response, the server returns administration system exception information that is related to the processing of the specified property.

```
[propget] HRESULT Exception(
    [out, retval] IAppHostPropertyException** ppException
);
```

ppException: Contains the exception information for a specified property.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppException is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.22.7 GetMetadata (Opnum 9)

The GetMetadata method is received by the server in an RPC_REQUEST packet. In response, the server returns the named metadata for the specified property.

```
HRESULT GetMetadata(
    [in] BSTR bstrMetadataType,
    [out, retval] VARIANT* pValue
);
```

bstrMetadataType: The name of the metadata property to fetch. Valid names are as follows.

Value	Meaning
"encryptionProvider"	A string that represents server-specific data that defines how the property will be

Value	Meaning
	encrypted or decrypted on the server. string
"isPropertyEncrypted"	A Boolean that determines if the property is encrypted. bool
"isDefaultValue"	A Boolean that represents whether the specified property was explicitly set in the administration system or is a system default. bool
"isInheritedFromDefault"	A Boolean that represents whether the specified property was explicitly set in the administration system or is a globally defined default. bool
"isLocked"	A Boolean that represents whether this property can be set at deeper hierarchy paths. bool

pValue: Contains the VARIANT value of the metadata property that is specified and the type depends on the property that is fetched.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property specified by bstrMetadataType is not supported.

3.1.4.22.8 SetMetadata (Opnum 10)

The SetMetadata method is received by the server in an RPC_REQUEST packet.

This method is used to set a metadata property.

```
HRESULT SetMetadata(
    [in] BSTR bstrMetadataType,
    [in] VARIANT value
);
```

bstrMetadataType: The name of the metadata to set on the property. Valid names are as follows.

Value	Meaning
"encryptionProvider"	A string that represents server-specific data that defines how the property will be

Value	Meaning
	encrypted or decrypted on the server. string

value: The value of the metadata property to set.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585 ERROR_INVALID_INDEX	The metadata property specified by bstrMetadataType is not supported.
0X80070013 ERROR_INVALID_DATA	The metadata property value has an unsupported type.
0X80070032 ERROR_NOT_SUPPORTED	The instance is not editable.

3.1.4.22.9 Schema (Opnum 11)

The Schema method is received by the server in an RPC_REQUEST packet. In response, the server returns the schema and constraints of the specified property, as defined in the [IAppHostPropertySchema](#) object that is returned.

```
[propget] HRESULT Schema(
    [out, retval] IAppHostPropertySchema** ppSchema
);
```

ppSchema: Set with the IAppHostPropertySchema for the corresponding [IAppHostProperty](#).

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSchema is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.23 IAppHostPropertyCollection

The IAppHostPropertyCollection **interface** provides methods that access a collection of [IAppHostProperty](#) objects that are supported by a corresponding [IAppHostElement](#).

The IAppHostPropertyCollection interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4

3.1.4.23.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the count of [IAppHostProperty](#) objects that are contained in the collection.

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcCount  
);
```

pcCount: Contains the count of the number of properties in the collection.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.23.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the property that is specified by the specific index.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT cIndex,  
    [out, retval] IAppHostProperty** ppProperty  
);
```

cIndex: The VARIANT index of the property to be fetched. If the VARIANT is of type integer, the index is a zero-based index to the collection. If the VARIANT is of type string, the index is a string that is the name of the property being retrieved.

ppProperty: Contains the retrieved property.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppProperty is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070585 ERROR_INVALID_INDEX	The integer value specified by cIndex is invalid, or there is no property with a name as specified by cIndex.

3.1.4.24 IAppHostPropertyException

The IAppHostPropertyException **interface** provides methods that access the exception information that the administration system encountered when processing the corresponding [IAppHostProperty](#). The administration system can indicate errors as encountered by filling in this exception. This behavior is defined by the [IAppHostAdminManager](#) metadata "ignoreInvalidAttributes".

The IAppHostPropertyException interface inherits opnums 0–9 from the [IAppHostConfigException](#) interface.

Methods in RPC Opnum Order

Method	Description
InvalidValue	Opnum: 10
ValidationFailureReason	Opnum: 11
ValidationFailureParameters	Opnum: 12

3.1.4.24.1 InvalidValue (Opnum 10)

The InvalidValue method is received by the server in an RPC_REQUEST packet. In response, the server returns a string representation of the invalid value that is encountered by the administration system when processing property.

```
[propget] HRESULT InvalidValue(  
    [out, retval] BSTR* pbstrValue
```

```
);
```

pbstrValue: Contains the invalid value representation.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrValue is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.24.2 ValidationFailureReason (Opnum 11)

The ValidationFailureReason method is received by the server in an RPC_REQUEST packet. In response, the server returns a description of the error that is encountered.

```
[propget] HRESULT ValidationFailureReason(  
    [out, retval] BSTR* pbstrValidationReason  
);
```

pbstrValidationReason: Contains a description of the error.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrValidationReason is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.24.3 ValidationFailureParameters (Opnum 12)

The ValidationFailureParameters method is currently reserved for future use. The server MUST return ERROR_NOT_SUPPORTED (as defined in [\[MS-ERREF\]](#)) to indicate that the method isn't implemented.

```
[propget] HRESULT ValidationFailureParameters(  
    [out, retval] SAFEARRAY (VARIANT)* pParameterArray  
);
```

pParameterArray: Contains the VARIANT array of parameters that are applicable to the error.

Return Values: The server MUST return ERROR_NOT_SUPPORTED (as defined in [\[MS-ERREF\]](#)) to indicate that the method isn't implemented.

3.1.4.25 IAppHostPropertySchema

The IAppHostPropertySchema **interface** provides methods that access the schema and constraints for the corresponding [IAppHostProperty](#) object.

The IAppHostPropertySchema interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Name	Opnum: 3
Type	Opnum: 4
DefaultValue	Opnum: 5
IsRequired	Opnum: 6
IsUniqueKey	Opnum: 7
IsCombinedKey	Opnum: 8
IsExpanded	Opnum: 9
ValidationType	Opnum: 10
ValidationParameter	Opnum: 11
GetMetadata	Opnum: 12
IsCaseSensitive	Opnum: 13
PossibleValues	Opnum: 14
DoesAllowInfinite	Opnum: 15
IsEncrypted	Opnum: 16
TimeSpanFormat	Opnum: 17

3.1.4.25.1 Name (Opnum 3)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server returns the name of the corresponding [IAppHostProperty](#).

```
[propget] HRESULT Name(
    [out, retval] BSTR* pbstrName
);
```

pbstrName: Contains the name of the corresponding property.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.25.2 Type (Opnum 4)

The Type method is received by the server in an RPC_REQUEST packet. In response, the server returns a string that represents the type of the property. Which types are supported is a server implementation detail.

```
[propget] HRESULT Type(
    [out, retval] BSTR* pbstrType
);
```

pbstrType: Contains the string that represents the type.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrType is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070013 ERROR_INVALID_DATA	The data is invalid. The attribute type is not supported.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.25.3 DefaultValue (Opnum 5)

The DefaultValue method is received by the server in an RPC_REQUEST packet. In response, the server returns the system-wide default value for the specified property, as defined by the administration system.

```
[propget] HRESULT DefaultValue(  
    [out, retval] VARIANT* pDefaultValue  
);
```

pDefaultValue: A VARIANT that contains the system default value.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070032 ERROR_NOT_SUPPORTED	The default value has a type that is not supported by the schema.

3.1.4.25.4 IsRequired (Opnum 6)

The IsRequired method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the specified property is required to be set on the server when the parent [IAppHostElement](#) exists.

```
[propget] HRESULT IsRequired(  
    [out, retval] VARIANT BOOL* pfIsRequired  
);
```

pfIsRequired: A Boolean value that states whether an attribute is required to be present in the parent IAppHostElement.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057	One or more parameters are incorrect or null.

Return value/code	Description
ERROR_INVALID_PARAMETER	

3.1.4.25.5 IsUniqueKey (Opnum 7)

The IsUniqueKey method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the specified property must be unique compared to all other properties of the peer collection of [IAppHostElement](#) objects. In other words, it applies only to properties that are members of the collection of IAppHostElement objects.

```
[propget] HRESULT IsUniqueKey(  
    [out, retval] VARIANT BOOL* pfIsUniqueKey  
);
```

pfIsUniqueKey: A Boolean value that states whether an attribute is required to be unique in the specified collection of the parent IAppHostElement.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.25.6 IsCombinedKey (Opnum 8)

The IsCombinedKey method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the specified property is part of a group of properties that combine to be unique compared to all other properties of peer collection [IAppHostElement](#) objects. In other words, it applies only to properties that are members of collection IAppHostElement objects.

```
[propget] HRESULT IsCombinedKey(  
    [out, retval] VARIANT BOOL* pfIsCombinedKey  
);
```

pfIsCombinedKey: A Boolean value that states whether an attribute is part of a combined key in the specified collection of the parent IAppHostElement object.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.25.7 IsExpanded (Opnum 9)

The IsExpanded method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the specified property supports being expanded on the server side to expand any embedded system environment variables.

```
[propget] HRESULT IsExpanded(
    [out, retval] VARIANT_BOOL* pfIsExpanded
);
```

pfIsExpanded: Set to whether the specified property supports environment variable expansion.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.25.8 ValidationType (Opnum 10)

The ValidationType method is received by the server in an RPC_REQUEST packet. In response, the server returns a string representing additional custom validation done when processing the corresponding property. The details of the validation are an implementation detail of the administration system.

```
[propget] HRESULT ValidationType(
    [out, retval] BSTR* pbstrValidationType
);
```

pbstrValidationType: Set to name of validation type performed on server.

Return Values: The server MUST return zero if it successfully processes the message received from the client. In this case, *pbstrValidationType is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.25.9 ValidationParameter (Opnum 11)

The ValidationParameter method is received by the server in an RPC_REQUEST packet. In response, the server returns any parameter that applies to the [ValidationType](#) of the specified property. Again, this is implementation-specific.

```
[propget] HRESULT ValidationParameter(
    [out, retval] BSTR* pbstrValidationParameter
);
```

pbstrValidationParameter: Set to the parameter of the validation type.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, pbstrValidationParameter is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.25.10 GetMetaData (Opnum 12)

The GetMetadata method is received by the server in an RPC_REQUEST packet. In response, the server returns any named metadata for the property schema.

No metadata is currently associated with this **interface**.

```
HRESULT GetMetadata(
    [in] BSTR bstrMetadataType,
    [out, retval] VARIANT* pValue
);
```

bstrMetadataType: The name of the metadata property to fetch.

pValue: Returns the value of the specified metadata property, and the type depends on the property fetched.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070032 ERROR_NOT_SUPPORTED	The metadata property specified by bstrMetadataType is not supported.

3.1.4.25.11 IsCaseSensitive (Opnum 13)

The IsCaseSensitive method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the corresponding property is compared to others in a case-sensitive manner, when determining equality for key (combined/unique) evaluation.

```
[propget] HRESULT IsCaseSensitive(  
    [out, retval] VARIANT_BOOL* pfIsCaseSensitive  
);
```

pfIsCaseSensitive: A Boolean value about whether the property is treated as case-sensitive.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.25.12 PossibleValues (Opnum 14)

The PossibleValues method is received by the server in an RPC_REQUEST packet. In response, the server returns a collection of the possible constant values for the specified property, if applicable. The administration system determines the applicability.

```
[propget] HRESULT PossibleValues(  

```

```
[out, retval] IAppHostConstantValueCollection** ppValues
);
```

ppValues: Contains the collection of possible values.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppValues is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.25.13 DoesAllowInfinite (Opnum 15)

The DoesAllowInfinite method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the property supports having an infinite value set.

```
[propget] HRESULT DoesAllowInfinite(
    [out, retval] VARIANT_BOOL* pfAllowInfinite
);
```

pfAllowInfinite: A pointer to a Boolean value that, if set to TRUE, indicates that the property that is represented by this [IAppHostPropertySchema](#) supports infinite values.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.25.14 IsEncrypted (Opnum 16)

The IsEncrypted method is received by the server in an RPC_REQUEST packet. In response, the server returns whether the corresponding [IAppHostProperty](#) will be encrypted when it is persisted in the administration system.

```
[propget] HRESULT IsEncrypted(
    [out, retval] VARIANT_BOOL* pfIsEncrypted
```

);

pfIsEncrypted: A pointer to a Boolean value that, if set to TRUE, indicates that the property that is represented by this [IAppHostPropertySchema](#) will be encrypted when persisted.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.25.15 TimeSpanFormat (Opnum 17)

The TimeSpanFormat method is received by the server in an RPC_REQUEST packet. In response, the server returns a format string that describes how the corresponding property is supposed to be formatted if the property represents a time span.

```
[propget] HRESULT TimeSpanFormat(  
    [out, retval] BSTR* pbstrTimeSpanFormat  
);
```

pbstrTimeSpanFormat: Contains the format string of the time span for the property.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrTimeSpanFormat is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.26 IAppHostPropertySchemaCollection

The IAppHostPropertySchemaCollection **interface** provides methods that access a collection of [IAppHostPropertySchema](#) objects.

The IAppHostPropertySchemaCollection interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4

3.1.4.26.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the count of the [IAppHostPropertySchema](#) objects in the specified collection.

```
[propget] HRESULT Count(  
    [out, retval] DWORD* pcCount  
);
```

pcCount: Contains the count of the IAppHostPropertySchema objects in the collection.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.26.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the [IAppHostPropertySchema](#) that is specified by the index.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT cIndex,  
    [out, retval] IAppHostPropertySchema** ppPropertySchema  
);
```

cIndex: A VARIANT that specifies the property schema to retrieve. If the VARIANT is of type integer, the index is the zero-based index to the collection. If the VARIANT is of type string, the index is the string name of the property schema to retrieve.

ppPropertySchema: Contains the IAppHostPropertySchema that is selected.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppPropertySchema is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by cIndex is invalid, or the IappHostPropertySchema instance with name specified by cIndex could not be found.

3.1.4.27 IAppHostSectionDefinition

The IAppHostSectionDefinition **interface** provides methods that access a declaration of the [IAppHostElement](#) object that is supported by the administration system. A declaration is distinct from the existence of an IAppHostElement in the administration system.

The IAppHostSectionDefinition interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Name	Opnum: 3
Type	"getter" Opnum: 4
Type	"setter" Opnum: 5
OverrideModeDefault	"getter" Opnum: 6
OverrideModeDefault	"setter" Opnum: 7
AllowDefinition	"getter" Opnum: 8
AllowDefinition	"setter" Opnum: 9
AllowLocation	"getter" Opnum: 10
AllowLocation	"setter"

Method	Description
	Opnum: 11

3.1.4.27.1 Name (Opnum 3)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server returns the name of the [IAppHostElement](#) being declared.

```
[propget] HRESULT Name(  
    [out, retval] BSTR* pbstrName  
);
```

pbstrName: The name of the IAppHostElement being declared.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.27.2 Type (Get) (Opnum 4)

The Type (Get) method is received by the server in an RPC_REQUEST packet. In response, the server returns a string that represents an implementation-specific type name for the declaration.

```
[propget] HRESULT Type(  
    [out, retval] BSTR* pbstrType  
);
```

pbstrType: Set to the type of the section.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrType is not NULL. If processing fails. the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000	The operation completed successfully.

Return value/code	Description
NO_ERROR	
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.27.3 Type (Set) (Opnum 5)

The Type (Set) method is received by the server in an RPC_REQUEST packet. In response, the server sets the type name of a specified declaration.

```
[propput] HRESULT Type(
    [in] BSTR bstrType
);
```

bstrType: The type name to set for the declaration.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.27.4 OverrideModeDefault (Get) (Opnum 6)

The OverrideModeDefault (Get) method is received by the server in an RPC_REQUEST packet. In response, the server returns an implementation-specific override behavior string for the declaration.

```
[propget] HRESULT OverrideModeDefault(
    [out, retval] BSTR* pbstrOverrideModeDefault
);
```

pbstrOverrideModeDefault: Contains the string that represents the override behavior.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, the *pbstrOverrideModeDefault* parameter is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.27.5 OverrideModeDefault (Set) (Opnum 7)

The OverrideModeDefault (Set) method is received by the server in an RPC_REQUEST packet. In response, the server sets an implementation-specific override behavior string for the declaration.

```
[propput] HRESULT OverrideModeDefault(
    [in] BSTR bstrOverrideModeDefault
);
```

bstrOverrideModeDefault: The override behavior string.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.27.6 AllowDefinition (Get) (Opnum 8)

The AllowDefinition (Get) method is received by the server in an RPC_REQUEST packet. In response, the server returns an implementation-specific string that defines where the specified declaration can apply in the administration system.

```
[propget] HRESULT AllowDefinition(
    [out, retval] BSTR* pbstrAllowDefinition
);
```

pbstrAllowDefinition: Contains the value that defines where the specified declaration can apply in the administration system.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrAllowDefinition is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.27.7 AllowDefinition (Set) (Opnum 9)

The AllowDefinition (Set) method is received by the server in an RPC_REQUEST packet. In response, the server sets an implementation-specific string that defines where the specified declaration can apply in the administration system.

```
[propput] HRESULT AllowDefinition(  
    [in] BSTR bstrAllowDefinition  
);
```

bstrAllowDefinition: A value that defines where the specified declaration can apply in the administration system.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.27.8 AllowLocation (Get) (Opnum 10)

The AllowLocation (Get) method is received by the server in an RPC_REQUEST packet. In response, the server returns an implementation-specific string that defines whether the declared [IAppHostElement](#) can exist in subpaths within an [IAppHostConfigFile](#).

```
[propget] HRESULT AllowLocation(  
    [out, retval] BSTR* pbstrAllowLocation  
);
```

pbstrAllowLocation: Contains the subpath behavior.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrAllowLocation is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.27.9 AllowLocation (Set) (Opnum 11)

The AllowLocation (Set) method is received by the server in an RPC_REQUEST packet. In response, the server sets an implementation-specific string that defines whether the declared [IAppHostElement](#) can exist in subpaths within an [IAppHostConfigFile](#).

```
[propput] HRESULT AllowLocation(  
    [in] BSTR bstrAllowLocation  
);
```

bstrAllowLocation: The subpath behavior to set.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.

3.1.4.28 IAppHostSectionDefinitionCollection

The IAppHostSectionDefinitionCollection **interface** provides methods that access a collection of [IAppHostSectionDefinition](#) objects.

The IAppHostSectionDefinitionCollection interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4
AddSection	Opnum: 5
DeleteSection	Opnum: 6

3.1.4.28.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns the count of the [IAppHostSectionDefinition](#) objects in the collection.

```
[propget] HRESULT Count(  
    [out, retval] unsigned long* pcCount  
);
```

pcCount: Contains the count of the IAppHostSectionDefinition objects.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.28.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns the [IAppHostSectionDefinition](#) object that is specified in the index.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT varIndex,  
    [out, retval] IAppHostSectionDefinition** ppConfigSection
```

);

varIndex: A VARIANT that specifies the section definition to retrieve. If the VARIANT is of type integer, the index is the zero-based index to the collection. If the VARIANT is of type string, the index is the string name of the section definition.

ppConfigSection: Contains the specified section definition.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppConfigSection is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by varIndex is invalid, or the element with name specified by varIndex could not be found.

3.1.4.28.3 AddSection (Opnum 5)

The AddSection method is received by the server in an RPC_REQUEST packet. In response, the server adds a section definition to the administration system.

```
HRESULT AddSection(  
    [in] BSTR bstrSectionName,  
    [out, retval] IAppHostSectionDefinition** ppConfigSection  
);
```

bstrSectionName: The name of the new section definition to add.

ppConfigSection: Contains the newly added or created section definition.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppConfigSection is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X800700B7	A section with name bstrSectionName already exists.

Return value/code	Description
ERROR_ALREADY_EXISTS	

3.1.4.28.4 DeleteSection (Opnum 6)

The DeleteSection method is received by the server in an RPC_REQUEST packet. In response, the server deletes the specified section definition.

```
HRESULT DeleteSection(
    [in] VARIANT varIndex
);
```

varIndex: A VARIANT index that specifies the section definition to delete. If the VARIANT is of type integer, the index is a zero-based index to the collection. If the VARIANT is of type string, the index is the string name of the section definition.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by varIndex is invalid, or the section with name specified by cIndex could not be found.
0X80070021 ERROR_LOCK_VIOLATION	The instance is set to read-only.
0X00000002 ERROR_PATH_NOT_FOUND	The system cannot find the path specified. The section could not be found.

3.1.4.29 IAppHostSectionGroup

The IAppHostSectionGroup **interface** provides methods that access a group of section definitions that have a common prefix name.

The IAppHostSectionGroup interface inherits opnums 0–2 from the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
Count	Opnum: 3
Item	Opnum: 4
Sections	Opnum: 5

Method	Description
AddSectionGroup	Opnum: 6
DeleteSectionGroup	Opnum: 7
Name	Opnum: 8
Type	"getter" Opnum: 9
Type	"setter" Opnum: 10

3.1.4.29.1 Count (Opnum 3)

The Count method is received by the server in an RPC_REQUEST packet. In response, the server returns a count of child section groups that are contained in the specified section group.

```
[propget] HRESULT Count(  
    [out, retval] unsigned long* pcSectionGroup  
);
```

pcSectionGroup: Contains the count of section groups in the collection array.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.29.2 Item (Opnum 4)

The Item method is received by the server in an RPC_REQUEST packet. In response, the server returns a section group that matches the specified index.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  
    [in] VARIANT varIndex,  
    [out, retval] IAppHostSectionGroup** ppSectionGroup  
);
```

varIndex: A VARIANT index that specifies the section group. If it is of type integer, the index is a zero-based index to the collection. If it is of type string, the index is the name of the section group to retrieve.

ppSectionGroup: Contains the selected [IAppHostSectionGroup](#).

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSectionGroup is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table below describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by varIndex is invalid, or the section group with name specified by varIndex could not be found.

3.1.4.29.3 Sections (Opnum 5)

The Sections method is received by the server in an RPC_REQUEST packet. In response, the server returns a collection of section definitions in the specified section group.

```
[propget] HRESULT Sections(
    [out, retval] IAppHostSectionDefinitionCollection** ppSections
);
```

ppSections: Contains the collection of section definitions.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSections is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

3.1.4.29.4 AddSectionGroup (Opnum 6)

The AddSectionGroup method is received by the server in an RPC_REQUEST packet. In response, the server adds a new section group to the specified section group.

```
HRESULT AddSectionGroup(
    [in] BSTR bstrSectionGroupName,
    [out, retval] IAppHostSectionGroup** ppSectionGroup
);
```

bstrSectionGroupName: A string that contains the name of the section group to add.

ppSectionGroup: Contains the pointer to the newly created section group.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *ppSectionGroup is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X800700B7 ERROR_ALREADY_EXISTS	A section group with name bstrSectionGroupName already exists.

3.1.4.29.5 DeleteSectionGroup (Opnum 7)

The DeleteSectionGroup method is received by the server in an RPC_REQUEST packet. In response, the server deletes the specified section group.

```
HRESULT DeleteSectionGroup(
    [in] VARIANT varIndex
);
```

varIndex: A VARIANT index that specifies the section group. If it is of type integer, the index is a zero-based index to the collection. If it is of type string, the index is the name of the section group to retrieve.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070585 ERROR_INVALID_INDEX	The integer index specified by varIndex is invalid, or the section group with name specified by cIndex could not be found.
0X80070021 ERROR_LOCK_VIOLATION	The instance is not editable.

3.1.4.29.6 Name (Opnum 8)

The Name method is received by the server in an RPC_REQUEST packet. In response, the server returns the name of the specified section group.

```
[propget] HRESULT Name(
    [out, retval] BSTR* pbstrName
);
```

pbstrName: Contains the string that contains the name of the section group.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrName is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.29.7 Type (Get) (Opnum 9)

The Type (Get) method is received by the server in an RPC_REQUEST packet. In response, the server returns an implementation-specific type string for the specified section group.

```
[propget] HRESULT Type(
    [out, retval] BSTR* pbstrType
);
```

pbstrType: Contains the string that contains the type string for the section group.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrType is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.29.8 Type (Set) (Opnum 10)

The Type (Set) method is received by the server in an RPC_REQUEST packet. In response, the server sets the implementation-specific type string for the specified section group.

```
[propput] HRESULT Type(  
    [in] BSTR bstrType  
);
```

bstrType: The string that contains the type of the section group.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.30 IAppHostWritableAdminManager

The IAppHostWritableAdminManager **interface** provides methods that access a writable version of an administration system. It extends the [IAppHostAdminManager](#), which is a read-only interface. The IAppHostWritableAdminManager adds methods to allow writing to the administration system, the most important of which is the CommitChanges method, which instructs the administration system to persist any in-memory changes that it has accumulated.

The IAppHostWritableAdminManager interface inherits opnums 0–6 from the IAppHostAdminManager interface, as defined in this protocol specification and the IUnknown interface.

Methods in RPC Opnum Order

Method	Description
CommitChanges	Opnum: 7
CommitPath	"getter" Opnum: 8
CommitPath	"setter" Opnum: 9

3.1.4.30.1 CommitChanges (Opnum 7)

The CommitChanges method is received by the server in an RPC_REQUEST packet. In response, the server commits any in-memory changes that it accumulates to a persisted store. This behavior essentially writes out the changes that are made by the client.

```
HRESULT CommitChanges();
```

This method has no parameters.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.
0X80070013 ERROR_INVALID_DATA	Configuration data or schema on the server are malformed or corrupted.
0X00000002 ERROR_PATH_NOT_FOUND	The system cannot find the path specified.
0X80070002 ERROR_FILE_NOT_FOUND	The system cannot find the file specified.
0X80070005 ERROR_ACCESS_DENIED	Access is denied.

3.1.4.30.2 CommitPath (Get) (Opnum 8)

The CommitPath (Get) method is received by the server in an RPC_REQUEST packet. In response, the server returns the hierarchy path that the administration system writes changes to after having its [CommitChanges](#) method called.

```
[propget] HRESULT CommitPath(  
    [out, retval] BSTR* pbstrCommitPath  
);
```

pbstrCommitPath: Contains the hierarchy path where changes are committed.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. In this case, *pbstrCommitPath is not NULL. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	One or more parameters are incorrect or null.

Return value/code	Description
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.4.30.3 CommitPath (Set) (Opnum 9)

The CommitPath (Set) method is received by the server in an RPC_REQUEST packet. In response, the server changes the hierarchy path where changes are committed.

```
[propput] HRESULT CommitPath(
    [in] BSTR bstrCommitPath
);
```

bstrCommitPath: The new hierarchy path to commit changes to.

Return Values: The server MUST return zero if it successfully processes the message that is received from the client. If processing fails, the server MUST return a nonzero HRESULT code as defined in [\[MS-ERREF\]](#). The following table describes the error conditions that MUST be handled and the corresponding error codes. A server MAY return additional implementation-specific error codes.

Return value/code	Description
0X00000000 NO_ERROR	The operation completed successfully.
0X80070057 ERROR_INVALID_PARAMETER	The parameter is incorrect. The commit path is invalid.
0x800700dd ERROR_FILE_CHECKED_OUT	The commit path cannot be set because there are some pending changes that have not been committed yet.
0X00000008 ERROR_NOT_ENOUGH_MEMORY	Not enough memory is available to process this command.

3.1.5 Timer Events

None.

3.1.6 Other Local Events

None.

4 Protocol Examples

This section shows examples of client use of this protocol. All examples are shown in JScript development software; however, any client scripting or programming language that can create and manipulate DCOM objects can be used.

For more information and sample usage of the protocol by a client, see [\[MSDN-IIS7AH\]](#).

4.1 Create an AppHostAdminManager Locally

```
var adminManager = WScript.CreateObject("Microsoft.ApplicationHost.AdminManager");
```

4.2 Get Metadata: Get the overrideMode of a defaultDocument Section

```
var adminManager = WScript.CreateObject("Microsoft.ApplicationHost.AdminManager");
var configSection = adminManager.GetAdminSection("system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST");
var overrideMode = configSection.GetMetadata("overrideMode");

/* WScript.Echo(overrideMode); */
```

4.3 Set Metadata: Set the overrideMode of the defaultDocument Section

```
var adminManager = WScript.CreateObject("Microsoft.ApplicationHost.WritableAdminManager");
adminManager.CommitPath = "MACHINE/WEBROOT/APPHOST";

var configSection = adminManager.GetAdminSection("system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST");
configSection.SetMetadata("overrideMode", "Deny");
adminManager.CommitChanges();
```

4.4 Create a New Configuration Section Entry in the configSections Section

```
var adminManager = WScript.CreateObject("Microsoft.ApplicationHost.WritableAdminManager");
adminManager.CommitPath = "MACHINE/WEBROOT/APPHOST";
var configFile = adminManager.ConfigManager.GetConfigFile("MACHINE/WEBROOT/APPHOST");

var sectionGroupTable = configFile.RootSectionGroup;

if ( sectionGroupTable != null )
{
    try
    {
        var sectionGroup = sectionGroupTable.Item("system.webServer");
    }
    catch ( exception )
    {
        var sectionGroup = null;
    }
}

if ( sectionGroup == null )
{
    sectionGroup=sectionGroupTable.AddSectionGroup("system.webServer");
}
```

```
var configSection=sectionGroup.Sections.AddSection("NewSectionGroup");
adminManager.CommitChanges();
```

4.5 Get a Section for Read Access: The defaultDocument Section

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.AdminManager" );
var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
```

4.6 Get a Property: Get the Enabled Property of the defaultDocument Section

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.AdminManager" );
var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
var isEnabled = configSection.Properties.Item( "enabled" ).Value;
```

4.7 Get a Section: Get the anonymousAuthentication Section

To get the anonymousAuthentication section:

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.AdminManager" );
var configSection = adminManager.GetAdminSection("system.webServer/security/authentication/anonymousAuthentication", "MACHINE/WEBROOT/APPHOST" );
```

To get a child element:

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.AdminManager" );
var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
var filesElement = configSection.GetElementByName( "files" );
```

4.8 List the Entries of a Collection

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.AdminManager" );
var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
var collection = configSection.GetElementByName( "files" ).Collection;

for ( var i = 0; i < collection.Count; i++ )
{
    WScript.Echo( "file name = " + collection.Item( i ).Properties.Item( "value" ).Value );
}
```

4.9 Remove an Entry of a Collection

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.WritableAdminManager" );
```

```
adminManager.CommitPath = "MACHINE/WEBROOT/APPHOST";

var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
var collection = configSection.GetElementByName( "files" ).Collection;
if ( collection.Count > 0 )
{
    collection.DeleteElement( 0 );
}
adminManager.CommitChanges();
```

4.10 Edit the Configuration of APPHOST in a Location Tag

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.WritableAdminManager" );
adminManager.CommitPath = "MACHINE/WEBROOT/APPHOST";

var configSection = adminManager.GetAdminSection(
    "system.webServer/urlCompression",
    "MACHINE/WEBROOT/APPHOST/Default Web Site" );
configSection.Properties.Item( "doDynamicCompression" ).Value = true;

adminManager.CommitChanges();
```

4.11 Read Schema Information: Determine If IsMergeAppend Is Set in the defaultDocuments Section

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.AdminManager" );
var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
var isMergeAppend = configSection.GetElementByName( "files" ).Collection.Schema.IsMergeAppend;
```

4.12 Get a Section for Write: Get the defaultDocument Section and Toggle the Enabled Attribute

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.WritableAdminManager" );
adminManager.CommitPath = "MACHINE/WEBROOT/APPHOST";

var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
configSection.Properties.Item( "enabled" ).Value = true;
adminManager.CommitChanges();
```

4.13 Write into a Collection: Clear the Contents of the defaultDocument Section for Site1

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.WritableAdminManager" );
adminManager.CommitPath = "MACHINE/WEBROOT/APPHOST";

var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
configSection.GetElementByName( "files" ).Collection.Clear();
```

```
adminManager.CommitChanges();
```

4.14 Write into a Collection: Add an Entry for the defaultDocument Section for Site1 as a Location Tag

```
var adminManager = WScript.CreateObject( "Microsoft.ApplicationHost.WritableAdminManager" );
adminManager.CommitPath = "MACHINE/WEBROOT/APPHOST";

var configSection = adminManager.GetAdminSection( "system.webServer/defaultDocument", "MACHINE/WEBROOT/APPHOST" );
var collection = configSection.GetElementByName( "files" ).Collection;

var newElement = collection.CreateNewElement( "add" );
newElement.Properties.Item( "value" ).Value = "newdefdoc.htm";

collection.AddElement( newElement );

adminManager.CommitChanges();
```

5 Security

5.1 Security Considerations for Implementers

Implementers have to enforce security as specified in [\[C706\]](#).

Implementers have to review the security considerations as specified in [\[MS-RPCE\]](#).

5.2 Index of Security Parameters

None.

6 Appendix A: Full IDL

For ease of implementation, the full **IDL** is provided where "ms-dtyp.idl" refers to the IDL found in [\[MS-DTYP\]](#), "ms-dcom.idl" is the IDL specified in [\[MS-DCOM\]](#), Appendix A, and "ms-oaut.idl" is the IDL that is specified in [\[MS-OAUT\]](#) Appendix A.

```
import "ms-dtyp.idl";
import "ms-dcom.idl";
import "ms-oaut.idl";

#define string

interface IAppHostMethod;
interface IAppHostMethodInstance;
interface IAppHostElement;
interface IAppHostProperty;
interface IAppHostConfigLocation;
interface IAppHostElementSchema;
interface IAppHostPropertySchema;
interface IAppHostConstantValue;
interface IAppHostConfigManager;

#define SAFEARRAY(x) SAFEARRAY

[
    object,
    uuid( 31a83ea0-c0e4-4a2c-8a01-353cc2a4c60a ),
    pointer default( unique ),
    helpstring( "IAppHostMappingExtension" )
]

interface IAppHostMappingExtension : IUnknown
{
    HRESULT GetSiteNameFromSiteId(
        [in] DWORD dwSiteId,
        [out, retval] BSTR * pbstrSiteName
    );

    HRESULT GetSiteIdFromSiteName(
        [in] BSTR bstrSiteName,
        [out, retval] DWORD * pdwSiteId
    );

    HRESULT GetSiteElementFromSiteId(
        [in] DWORD dwSiteId,
        [out, retval] IAppHostElement ** ppSiteElement
    );

    HRESULT MapPath(
        [in] BSTR bstrSiteName,
        [in] BSTR bstrVirtualPath,
        [out] BSTR * pbstrPhysicalPath,
        [out] IAppHostElement ** ppVirtualDirectoryElement,
        [out] IAppHostElement ** ppApplicationElement
    );
};

[
    object,
    uuid( 08a90f5f-0702-48d6-b45f-02a9885a9768 ),
    pointer default( unique ),
    helpstring( "IAppHostChildElementCollection" )
]
interface IAppHostChildElementCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] DWORD * pcCount
    );
};
```

```

    );

    [propget, id(DISPID_VALUE)] HRESULT Item(
        [in] VARIANT cIndex,
        [out, retval] IAppHostElement ** ppElement
    );
}

[
    object,
    uuid( 0191775e-bcff-445a-b4f4-3bdda54e2816 ),
    pointer_default( unique ),
    helpstring( "IAppHostPropertyCollection" )
]

interface IAppHostPropertyCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] DWORD * pcCount
    );

    [propget, id(DISPID_VALUE)] HRESULT Item(
        [in] VARIANT cIndex,
        [out, retval] IAppHostProperty ** ppProperty
    );
}

[
    object,
    uuid( 832a32f7-b3ea-4b8c-b260-9a2923001184 ),
    pointer default( unique ),
    helpstring( "IAppHostConfigLocationCollection" )
]

interface IAppHostConfigLocationCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] DWORD * pcCount
    );

    [propget, id(DISPID_VALUE)] HRESULT Item(
        [in] VARIANT varIndex,
        [out, retval] IAppHostConfigLocation ** ppLocation
    );

    HRESULT AddLocation(
        [in] BSTR bstrLocationPath,
        [out, retval] IAppHostConfigLocation ** ppNewLocation
    );

    HRESULT DeleteLocation(
        [in] VARIANT cIndex
    );
}

[
    object,
    uuid( d6c7cd8f-bb8d-4f96-b591-d3a5f1320269 ),
    pointer default( unique ),
    helpstring( "IAppHostMethodCollection" )
]

interface IAppHostMethodCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] DWORD * pcCount
    );

    [propget, id(DISPID_VALUE)] HRESULT Item(

```

```

        [in] VARIANT cIndex,
        [out, retval] IAppHostMethod ** ppMethod
    );
}

[
    object,
    uuid( 0344cdda-151e-4cbf-82da-66ae61e97754 ),
    pointer_default( unique ),
    helpstring( "IAppHostElementSchemaCollection" )
]

interface IAppHostElementSchemaCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] DWORD * pcCount
    );

    [propget, id(DISPID_VALUE)] HRESULT Item(
        [in] VARIANT cIndex,
        [out, retval] IAppHostElementSchema ** ppElementSchema
    );
}

[
    object,
    uuid( 8bed2c68-a5fb-4b28-8581-a0dc5267419f ),
    pointer_default( unique ),
    helpstring( "IAppHostPropertySchemaCollection" )
]

interface IAppHostPropertySchemaCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] DWORD * pcCount
    );

    [propget, id(DISPID_VALUE)] HRESULT Item(
        [in] VARIANT cIndex,
        [out, retval] IAppHostPropertySchema ** ppPropertySchema
    );
}

[
    object,
    uuid( 5b5a68e6-8b9f-45e1-8199-a95ffccdfcff ),
    pointer_default( unique ),
    helpstring( "IAppHostConstantValueCollection" )
]

interface IAppHostConstantValueCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] DWORD * pcCount
    );

    [propget, id(DISPID_VALUE)] HRESULT Item(
        [in] VARIANT cIndex,
        [out, retval] IAppHostConstantValue ** ppConstantValue
    );
}

[
    object,
    uuid( 0716caf8-7d05-4a46-8099-77594be91394 ),
    pointer_default( unique ),
    helpstring( "IAppHostConstantValue" )
]

```



```

interface IAppHostConstantValue : IUnknown
{
    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );

    [propget] HRESULT Value(
        [out, retval] DWORD * pdwValue
    );
}

[
    object,
    uuid( 450386db-7409-4667-935e-384dbbee2a9e ),
    pointer_default( unique ),
    helpstring( "IAppHostPropertySchema" )
]

interface IAppHostPropertySchema : IUnknown
{
    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );

    [propget] HRESULT Type(
        [out, retval] BSTR * pbstrType
    );

    [propget] HRESULT DefaultValue(
        [out, retval] VARIANT * pDefaultValue
    );

    [propget] HRESULT IsRequired(
        [out, retval] VARIANT_BOOL * pfIsRequired
    );

    [propget] HRESULT IsUniqueKey(
        [out, retval] VARIANT_BOOL * pfIsUniqueKey
    );

    [propget] HRESULT IsCombinedKey(
        [out, retval] VARIANT_BOOL * pfIsCombinedKey
    );

    [propget] HRESULT IsExpanded(
        [out, retval] VARIANT_BOOL * pfIsExpanded
    );

    [propget] HRESULT ValidationType(
        [out, retval] BSTR * pbstrValidationType
    );

    [propget] HRESULT ValidationParameter(
        [out, retval] BSTR * pbstrValidationParameter
    );

    HRESULT GetMetadata(
        [in] BSTR bstrMetadataType,
        [out, retval] VARIANT * pValue
    );

    [propget] HRESULT IsCaseSensitive(
        [out, retval] VARIANT_BOOL * pfIsCaseSensitive
    );

    [propget] HRESULT PossibleValues(
        [out, retval] IAppHostConstantValueCollection ** ppValues
    );

    [propget] HRESULT DoesAllowInfinite(

```

```

        [out, retval] VARIANT_BOOL * pfAllowInfinite
    );

    [propget] HRESULT IsEncrypted(
        [out, retval] VARIANT_BOOL * pfIsEncrypted
    );

    [propget] HRESULT TimeSpanFormat(
        [out, retval] BSTR * pbstrTimeSpanFormat
    );
}

[
    object,
    uuid( de095db1-5368-4d11-81f6-efef619b7bcf ),
    pointer_default( unique ),
    helpstring( "IAppHostCollectionSchema" )
]

interface IAppHostCollectionSchema : IUnknown
{
    [propget] HRESULT AddElementNames(
        [out, retval] BSTR * pbstrElementName
    );

    HRESULT GetAddElementSchema(
        [in] BSTR bstrElementName,
        [out, retval] IAppHostElementSchema ** ppSchema
    );

    [propget] HRESULT RemoveElementSchema(
        [out, retval] IAppHostElementSchema ** ppSchema
    );

    [propget] HRESULT ClearElementSchema(
        [out, retval] IAppHostElementSchema ** ppSchema
    );

    [propget] HRESULT IsMergeAppend(
        [out, retval] VARIANT_BOOL * pfIsMergeAppend
    );
    HRESULT GetMetadata(
        [in] BSTR bstrMetadataType,
        [out, retval] VARIANT * pValue
    );

    [propget] HRESULT DoesAllowDuplicates(
        [out, retval] VARIANT_BOOL * pfAllowDuplicates
    );
}

[
    object,
    uuid( ef13d885-642c-4709-99ec-b89561c6bc69 ),
    pointer_default( unique ),
    helpstring( "IAppHostElementSchema" )
]

interface IAppHostElementSchema : IUnknown
{
    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );

    [propget] HRESULT DoesAllowUnschemasitizedProperties(
        [out, retval] VARIANT_BOOL * pfAllowUnschemasitized
    );

    HRESULT GetMetadata(

```

```

        [in] BSTR bstrMetadataType,
        [out, retval] VARIANT * pValue
    );

    [propget] HRESULT CollectionSchema(
        [out, retval] IAppHostCollectionSchema ** ppCollectionSchema
    );

    [propget] HRESULT ChildElementSchemas(
        [out, retval] IAppHostElementSchemaCollection ** ppChildSchemas
    );

    [propget] HRESULT PropertySchemas(
        [out, retval] IAppHostPropertySchemaCollection ** ppPropertySchemas
    );

    [propget] HRESULT IsCollectionDefault(
        [out, retval] VARIANT_BOOL * pfIsCollectionDefault
    );
}

[
    object,
    uuid( 2d9915fb-9d42-4328-b782-1b46819fab9e ),
    pointer_default( unique ),
    helpstring( "IAppHostMethodSchema" )
]

interface IAppHostMethodSchema : IUnknown
{
    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );

    [propget] HRESULT InputSchema(
        [out, retval] IAppHostElementSchema ** ppInputSchema
    );

    [propget] HRESULT OutputSchema(
        [out, retval] IAppHostElementSchema ** ppOutputSchema
    );

    HRESULT GetMetadata(
        [in] BSTR bstrMetadataType,
        [out, retval] VARIANT * pValue
    );
}

[
    object,
    uuid( b80f3c42-60e0-4ae0-9007-f52852d3dbed ),
    pointer default( unique ),
    helpstring( "IAppHostMethodInstance" )
]

interface IAppHostMethodInstance : IUnknown
{
    [propget] HRESULT Input(
        [out, retval] IAppHostElement ** ppInputElement
    );

    [propget] HRESULT Output(
        [out, retval] IAppHostElement ** ppOutputElement
    );

    HRESULT Execute();

    HRESULT GetMetadata(
        [in] BSTR bstrMetadataType,

```

```

        [out, retval] VARIANT * pValue
    );

    HRESULT SetMetadata(
        [in] BSTR bstrMetadataType,
        [in] VARIANT value
    );
}

[
    object,
    uuid( 7883calc-1112-4447-84c3-52fbeb38069d ),
    pointer_default( unique ),
    helpstring( "IAppHostMethod" )
]

interface IAppHostMethod : IUnknown
{
    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );

    [propget] HRESULT Schema(
        [out, retval] IAppHostMethodSchema ** ppMethodSchema
    );

    HRESULT CreateInstance(
        [out, retval] IAppHostMethodInstance ** ppMethodInstance
    );
}

[
    object,
    uuid( 4dfaldf3-8900-4bc7-bbb5-d1a458c52410 ),
    pointer_default( unique ),
    helpstring( "IAppHostConfigException" )
]

interface IAppHostConfigException : IUnknown
{
    [propget] HRESULT LineNumber(
        [out, retval] unsigned long * pcLineNumber
    );

    [propget] HRESULT FileName(
        [out, retval] BSTR * pbstrFileName
    );

    [propget] HRESULT ConfigPath(
        [out, retval] BSTR * pbstrConfigPath
    );

    [propget] HRESULT ErrorLine(
        [out, retval] BSTR * pbstrErrorLine
    );

    [propget] HRESULT PreErrorLine(
        [out, retval] BSTR * pbstrPreErrorLine
    );

    [propget] HRESULT PostErrorLine(
        [out, retval] BSTR * pbstrPostErrorLine
    );

    [propget] HRESULT ErrorString(
        [out, retval] BSTR * pbstrErrorString
    );
}

```

```

[
    object,
    uuid( eafe4895-a929-41ea-b14d-613e23f62b71 ),
    pointer_default( unique ),
    helpstring( "IAppHostPropertyException" )
]

interface IAppHostPropertyException : IAppHostConfigException
{
    [propget] HRESULT InvalidValue(
        [out, retval] BSTR * pbstrValue
    );

    [propget] HRESULT ValidationFailureReason(
        [out, retval] BSTR * pbstrValidationReason
    );

    [propget] HRESULT ValidationFailureParameters(
        [out, retval] SAFEARRAY(VARIANT) * pParameterArray
    );
}

[
    object,
    uuid( c8550bff-5281-4b1e-ac34-99b6fa38464d ),
    pointer_default( unique ),
    helpstring( "IAppHostElementCollection" )
]

interface IAppHostElementCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] DWORD * pcElementCount
    );

    [propget, id(DISPID VALUE)] HRESULT Item(
        [in] VARIANT cIndex,
        [out, retval] IAppHostElement ** ppElement
    );

    HRESULT AddElement(
        [in] IAppHostElement * pElement,
        [in, defaultvalue(-1)] int cPosition
    );

    HRESULT DeleteElement(
        [in] VARIANT cIndex
    );

    HRESULT Clear();

    HRESULT CreateNewElement(
        [in, defaultvalue("")] BSTR bstrElementName,
        [out, retval] IAppHostElement ** ppElement
    );

    [propget] HRESULT Schema(
        [out, retval] IAppHostCollectionSchema** ppSchema
    );
}

[
    object,
    uuid( 64ff8ccc-b287-4dae-b08a-a72cbf45f453 ),
    pointer_default( unique ),
    helpstring( "IAppHostElement" )
]

interface IAppHostElement : IUnknown

```

```

{
    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );

    [propget] HRESULT Collection(
        [out, retval] IAppHostElementCollection ** ppCollection
    );

    [propget] HRESULT Properties(
        [out, retval] IAppHostPropertyCollection ** ppProperties
    );

    [propget] HRESULT ChildElements(
        [out, retval] IAppHostChildElementCollection ** ppElements
    );
    HRESULT GetMetadata(
        [in] BSTR bstrMetadataType,
        [out, retval] VARIANT * pValue
    );

    HRESULT SetMetadata(
        [in] BSTR bstrMetadataType,
        [in] VARIANT value
    );

    [propget] HRESULT Schema(
        [out, retval] IAppHostElementSchema ** ppSchema
    );

    HRESULT GetElementByName(
        [in] BSTR bstrSubName,
        [out, retval] IAppHostElement ** ppElement
    );

    HRESULT GetPropertyByName(
        [in] BSTR bstrSubName,
        [out, retval] IAppHostProperty ** ppProperty
    );

    HRESULT Clear();

    [propget] HRESULT Methods(
        [out, retval] IAppHostMethodCollection ** ppMethods
    );
}

[
    object,
    uuid( ed35f7a1-5024-4e7b-a44d-07ddaf4b524d ),
    pointer_default( unique ),
    helpstring( "IAppHostProperty" )
]

interface IAppHostProperty : IUnknown
{
    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );

    [propget] HRESULT Value(
        [out, retval] VARIANT * pVariant
    );

    [propput] HRESULT Value(
        [in] VARIANT value
    );

    HRESULT Clear();
}

```

```

    [propget] HRESULT StringValue(
        [out, retval] BSTR * pbstrValue
    );

    [propget] HRESULT Exception(
        [out, retval] IAppHostPropertyException ** ppException
    );

    HRESULT GetMetadata(
        [in] BSTR bstrMetadataType,
        [out, retval] VARIANT * pValue
    );

    HRESULT SetMetadata(
        [in] BSTR bstrMetadataType,
        [in] VARIANT value
    );

    [propget] HRESULT Schema(
        [out, retval] IAppHostPropertySchema ** ppSchema
    );
}

[
    object,
    uuid( 370af178-7758-4dad-8146-7391f6e18585 ),
    pointer_default( unique ),
    helpstring( "IAppHostConfigLocation" )
]

interface IAppHostConfigLocation : IUnknown
{
    [propget]
    HRESULT Path(
        [out, retval] BSTR * pbstrLocationPath
    );

    [propget] HRESULT Count(
        [out, retval] DWORD * pcCount
    );

    [propget, id(DISPID_VALUE)] HRESULT Item(
        [in] VARIANT cIndex,
        [out, retval] IAppHostElement ** ppSection
    );

    HRESULT AddConfigSection(
        [in] BSTR bstrSectionName,
        [out, retval] IAppHostElement ** ppAdminElement
    );

    HRESULT DeleteConfigSection(
        [in] VARIANT cIndex
    );
}

[
    object,
    uuid( c5c04795-321c-4014-8fd6-d44658799393 ),
    pointer_default( unique ),
    helpstring( "IAppHostSectionDefinition" )
]

interface IAppHostSectionDefinition : IUnknown
{
    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );
}

```

```

    [propget] HRESULT Type(
        [out, retval] BSTR * pbstrType
    );

    [propput] HRESULT Type(
        [in] BSTR bstrType
    );

    [propget] HRESULT OverrideModeDefault(
        [out, retval] BSTR * pbstrOverrideModeDefault
    );

    [propput] HRESULT OverrideModeDefault(
        [in] BSTR bstrOverrideModeDefault
    );

    [propget] HRESULT AllowDefinition(
        [out, retval] BSTR * pbstrAllowDefinition
    );

    [propput] HRESULT AllowDefinition(
        [in] BSTR bstrAllowDefinition
    );

    [propget] HRESULT AllowLocation(
        [out, retval] BSTR * pbstrAllowLocation
    );

    [propput] HRESULT AllowLocation(
        [in] BSTR bstrAllowLocation
    );
}

[
    object,
    uuid( b7d381ee-8860-47a1-8af4-1f33b2b1f325 ),
    pointer_default( unique ),
    helpstring( "IAppHostSectionDefinitionCollection" )
]

interface IAppHostSectionDefinitionCollection : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] unsigned long * pcCount
    );

    [propget, id(DISPID VALUE)] HRESULT Item(
        [in] VARIANT varIndex,
        [out, retval] IAppHostSectionDefinition ** ppConfigSection
    );

    HRESULT AddSection(
        [in] BSTR bstrSectionName,
        [out, retval] IAppHostSectionDefinition ** ppConfigSection
    );

    HRESULT DeleteSection(
        [in] VARIANT varIndex
    );
}

[
    object,
    uuid( 0dd8a158-ebe6-4008-a1d9-b7ecc8f1104b ),
    pointer_default( unique ),
    helpstring( "IAppHostSectionGroup" )
]

```



```

interface IAppHostSectionGroup : IUnknown
{
    [propget] HRESULT Count(
        [out, retval] unsigned long * pcSectionGroup
    );

    [propget, id(DISPID_VALUE)] HRESULT Item(
        [in] VARIANT varIndex,
        [out, retval] IAppHostSectionGroup ** ppSectionGroup
    );

    [propget] HRESULT Sections(
        [out, retval] IAppHostSectionDefinitionCollection ** ppSections
    );

    HRESULT AddSectionGroup(
        [in] BSTR bstrSectionGroupName,
        [out, retval] IAppHostSectionGroup ** ppSectionGroup
    );
    HRESULT DeleteSectionGroup(
        [in] VARIANT varIndex
    );

    [propget] HRESULT Name(
        [out, retval] BSTR * pbstrName
    );

    [propget] HRESULT Type(
        [out, retval] BSTR * pbstrType
    );

    [propput] HRESULT Type(
        [in] BSTR bstrType
    );
}

[
    object,
    uuid( ada4e6fb-e025-401e-a5d0-c3134a281f07 ),
    pointer default( unique ),
    helpstring( "IAppHostConfigFile" )
]

interface IAppHostConfigFile : IUnknown
{
    [propget] HRESULT ConfigPath(
        [out, retval] BSTR * pbstrConfigPath
    );

    [propget] HRESULT FilePath(
        [out, retval] BSTR * pbstrFilePath
    );

    [propget] HRESULT Locations(
        [out, retval] IAppHostConfigLocationCollection ** ppLocations
    );

    HRESULT GetAdminSection(
        [in] BSTR bstrSectionName,
        [in] BSTR bstrPath,
        [out, retval] IAppHostElement ** ppAdminSection
    );

    HRESULT GetMetadata(
        [in] BSTR bstrMetadataType,
        [out, retval] VARIANT * pValue
    );

    HRESULT SetMetadata(

```

```

        [in] BSTR bstrMetadataType,
        [in] VARIANT value
    );

    HRESULT ClearInvalidSections();

    [propget] HRESULT RootSectionGroup(
        [out, retval] IAppHostSectionGroup ** ppSectionGroups
    );
}

[
    object,
    uuid( e7927575-5cc3-403b-822e-328a6b904bee ),
    pointer_default( unique ),
    helpstring( "IAppHostPathMapper" )
]

interface IAppHostPathMapper : IUnknown
{
    HRESULT MapPath(
        [in] BSTR bstrConfigPath,
        [in] BSTR bstrMappedPhysicalPath,
        [out, retval] BSTR * pbstrNewPhysicalPath
    );
}

[
    object,
    uuid( 09829352-87c2-418d-8d79-4133969a489d ),
    pointer default( unique ),
    helpstring( "IAppHostChangeHandler" )
]

interface IAppHostChangeHandler : IUnknown
{
    HRESULT OnSectionChanges(
        [in] BSTR bstrSectionName,
        [in] BSTR bstrConfigPath
    );
}

[
    object,
    uuid( 9be77978-73ed-4a9a-87fd-13f09fec1b13 ),
    pointer default( unique ),
    helpstring( "IAppHostAdminManager Interface" )
]

interface IAppHostAdminManager : IUnknown
{
    HRESULT GetAdminSection(
        [in] BSTR bstrSectionName,
        [in] BSTR bstrPath,
        [out, retval] IAppHostElement ** ppAdminSection
    );

    HRESULT GetMetadata(
        [in] BSTR bstrMetadataType,
        [out, retval] VARIANT * pValue
    );

    HRESULT SetMetadata(
        [in] BSTR bstrMetadataType,
        [in] VARIANT value
    );

    [propget] HRESULT ConfigManager(
        [out, retval] IAppHostConfigManager ** ppConfigManager
    );
}

```

```

    );
}

[
    object,
    uuid( fa7660f6-7b3f-4237-a8bf-ed0ad0dcbbd9 ),
    pointer_default( unique ),
    helpstring( "IAppHostWritableAdminManager Interface" )
]

interface IAppHostWritableAdminManager : IAppHostAdminManager
{
    HRESULT CommitChanges();

    [propget] HRESULT CommitPath(
        [out, retval] BSTR * pbstrCommitPath
    );

    [propput] HRESULT CommitPath(
        [in] BSTR bstrCommitPath
    );
}

[
    object,
    uuid( 8f6d760f-f0cb-4d69-b5f6-848b33e9bdc6 ),
    pointer_default( unique ),
    helpstring( "IAppHostConfigManager Interface" )
]

interface IAppHostConfigManager : IUnknown
{
    HRESULT GetConfigFile(
        [in] BSTR bstrConfigPath,
        [out, retval] IAppHostConfigFile ** ppConfigFile
    );

    HRESULT GetUniqueConfigPath(
        [in] BSTR bstrConfigPath,
        [out, retval] BSTR * pbstrUniquePath
    );
}

```

7 Appendix B: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

Windows Releases

- Windows Vista operating system with Service Pack 1 (SP1)
- Windows Server 2008 operating system
- Windows 7 operating system
- Windows Server 2008 R2 operating system
- Windows 8 operating system
- Windows Server 2012 operating system
- Windows 8.1 operating system
- Windows Server 2012 R2 operating system
- Windows 10 operating system
- Windows Server 2016 operating system
- Windows Server operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows 11 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 1.9](#): Windows supports these values.

8 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
Z Appendix B: Product Behavior	Added Windows Server 2025 to the list of applicable products.	Major

9 Index

A

Abstract data model
 [server](#) 15
[Abstract data model - server](#) 15
[AddConfigSection method](#) 42
[AddElement method](#) 63
[AddElementNames method](#) 27
[AddLocation method](#) 45
[AddSection method](#) 112
[AddSectionGroup method](#) 115
AllowDefinition method ([section 3.1.4.27.6](#) 108,
 [section 3.1.4.27.7](#) 109)
AllowLocation method ([section 3.1.4.27.8](#) 109,
 [section 3.1.4.27.9](#) 110)
[Applicability](#) 11

C

[Capability negotiation](#) 11
[Change tracking](#) 141
[ChildElements method](#) 54
[ChildElementSchemas method](#) 69
Clear method ([section 3.1.4.12.10](#) 60, [section 3.1.4.13.5](#) 64, [section 3.1.4.22.4](#) 87)
[ClearElementSchema method](#) 29
[ClearInvalidSections method](#) 39
[Collection method](#) 53
[CollectionSchema method](#) 68
[CommitChanges method](#) 118
CommitPath method ([section 3.1.4.30.2](#) 119, [section 3.1.4.30.3](#) 119)
[Common data types](#) 14
[ConfigManager method](#) 24
ConfigPath method ([section 3.1.4.5.3](#) 32, [section 3.1.4.6.1](#) 36)
Count method ([section 3.1.4.3.1](#) 25, [section 3.1.4.7.2](#) 41, [section 3.1.4.8.1](#) 44, [section 3.1.4.11.1](#) 50, [section 3.1.4.13.1](#) 62, [section 3.1.4.15.1](#) 71, [section 3.1.4.18.1](#) 77, [section 3.1.4.23.1](#) 92, [section 3.1.4.26.1](#) 104, [section 3.1.4.28.1](#) 111, [section 3.1.4.29.1](#) 113)
[Create a new configuration section entry in the configsections section example](#) 121
[Create an apphostadminmanager locally example](#) 121
[CreateInstance method](#) 77
[CreateNewElement method](#) 65
[Creating AppHostAdminManager locally - example](#) 121
[Creating configuration section entry - example](#) 121

D

Data model - abstract
 [server](#) 15
[Data model - abstract - server](#) 15
[Data types](#) 14
 [common - overview](#) 14
[DefaultValue method](#) 96
[DeleteConfigSection method](#) 43

[DeleteElement method](#) 63
[DeleteLocation method](#) 46
[DeleteSection method](#) 112
[DeleteSectionGroup method](#) 116
[DoesAllowDuplicates method](#) 30
[DoesAllowInfinite method](#) 102
[DoesAllowUnschemasizedProperties method](#) 67

E

[Edit the configuration of apphost in a location tag example](#) 123
[Editing APPHOST configuration in location tag - example](#) 123
[ErrorLine method](#) 33
[ErrorString method](#) 35
Events
 [local - server](#) 120
 [timer - server](#) 120
Examples
 [create a new configuration section entry in the configsections section](#) 121
 [create an apphostadminmanager locally](#) 121
 [creating AppHostAdminManager locally](#) 121
 [creating configuration section entry](#) 121
 [edit the configuration of apphost in a location tag](#) 123
 [editing APPHOST configuration in location tag](#) 123
 [get a property: get the enabled property of the defaultdocument section](#) 122
 [get a section for read access: the defaultdocument section](#) 122
 [get a section for write: get the defaultdocument section and toggle the enabled attribute](#) 123
 [get a section: get the anonymousauthentication section](#) 122
 [get metadata: get the override mode of a defaultdocument section](#) 121
 [getting metadata](#) 121
 [getting property](#) 122
 [getting section](#) 122
 [getting section for read access](#) 122
 [getting section for write](#) 123
 [list the entries of a collection](#) 122
 [listing entries of collection](#) 122
 [overview](#) 121
 [read schema information: determine if ismergeappend is set in the defaultdocuments section](#) 123
 [reading schema information](#) 123
 [remove an entry of a collection](#) 122
 [removing entry of collection](#) 122
 [set metadata: set the override mode of the defaultdocument section](#) 121
 [setting metadata](#) 121
 [write into a collection: add an entry for the defaultdocument section for site1 as a location tag](#) 124
 [write into a collection: clear the contents of the defaultdocument section for site1](#) 123
 writing into collection

[adding entry as location tag](#) 124
[clearing contents](#) 123
[Exception method](#) 88
[Execute method](#) 80

F

[Fields - vendor-extensible](#) 12
[FileName method](#) 32
[FilePath method](#) 36
[Full IDL](#) 126

G

[Get a property: get the enabled property of the defaultdocument section example](#) 122
[Get a section for read access: the defaultdocument section example](#) 122
[Get a section for write: get the defaultdocument section and toggle the enabled attribute example](#) 123
[Get a section: get the anonymousauthentication section example](#) 122
[Get metadata: get the override mode of a defaultdocument section example](#) 121
[GetAddElementSchema method](#) 28
[GetAdminSection method \(section 3.1.4.1.1 19, section 3.1.4.6.4 37\)](#)
[GetConfigFile method](#) 47
[GetElementByName method](#) 59
[GetMetadata method \(section 3.1.4.1.2 20, section 3.1.4.4.6 30, section 3.1.4.6.5 38, section 3.1.4.12.5 55, section 3.1.4.14.3 67, section 3.1.4.19.4 80, section 3.1.4.20.4 83, section 3.1.4.22.7 89, section 3.1.4.25.10 100\)](#)
[GetPropertyByName method](#) 59
[GetSiteElementFromSiteId method](#) 74
[GetSiteIdFromSiteName method](#) 73
[GetSiteNameFromSiteId method](#) 72
[Getting metadata example](#) 121
[Getting property example](#) 122
[Getting section example](#) 122
[Getting section for read access example](#) 122
[Getting section for write example](#) 123
[GetUniqueConfigPath method](#) 48
[Glossary](#) 10

I

[IAppHostAdminManager method](#) 18
[IAppHostChangeHandler method](#) 24
[IAppHostChildElementCollection method](#) 25
[IAppHostCollectionSchema method](#) 27
[IAppHostConfigException method](#) 31
[IAppHostConfigFile method](#) 35
[IAppHostConfigLocationCollection method](#) 40
[IAppHostConfigLocationCollection method](#) 44
[IAppHostConfigManager method](#) 47
[IAppHostConstantValue method](#) 49
[IAppHostConstantValueCollection method](#) 50
[IAppHostElement method](#) 52
[IAppHostElementCollection method](#) 61
[IAppHostElementSchema method](#) 66
[IAppHostElementSchemaCollection method](#) 70
[IAppHostMappingExtension method](#) 72

[IAppHostMethod method](#) 75
[IAppHostMethodCollection method](#) 77
[IAppHostMethodInstance method](#) 79
[IAppHostMethodSchema method](#) 82
[IAppHostPathMapper method](#) 84
[IAppHostProperty method](#) 85
[IAppHostPropertyCollection method](#) 91
[IAppHostPropertyException method](#) 93
[IAppHostPropertySchema method](#) 95
[IAppHostPropertySchemaCollection method](#) 103
[IAppHostSectionDefinition method](#) 105
[IAppHostSectionDefinitionCollection method](#) 110
[IAppHostSectionGroup method](#) 113
[IAppHostWritableAdminManager method](#) 118
[IDL](#) 126
[Implementer - security considerations](#) 125
[Index of security parameters](#) 125
[Informative references](#) 11
[Initialization](#)
[server](#) 18
[Initialization - server](#) 18
[Input method](#) 79
[InputSchema method](#) 83
[Introduction](#) 10
[InvalidValue method](#) 93
[IsCaseSensitive method](#) 101
[IsCollectionDefault method](#) 70
[IsCombinedKey method](#) 98
[IsEncrypted method](#) 102
[IsExpanded method](#) 98
[IsMergeAppend method](#) 30
[IsRequired method](#) 97
[IsUniqueKey method](#) 97
[Item method \(section 3.1.4.3.2 26, section 3.1.4.7.3 42, section 3.1.4.8.2 45, section 3.1.4.11.2 51, section 3.1.4.13.2 62, section 3.1.4.15.2 71, section 3.1.4.18.2 78, section 3.1.4.23.2 92, section 3.1.4.26.2 104, section 3.1.4.28.2 111, section 3.1.4.29.2 114\)](#)

L

[LineNumber method](#) 31
[List the entries of a collection example](#) 122
[Listing entries of collection - example](#) 122
[Local events](#)
[server](#) 120
[Local events - server](#) 120
[Locations method](#) 37

M

[MapPath method \(section 3.1.4.16.4 74, section 3.1.4.21.1 84\)](#)
[Message processing](#)
[server](#) 18
[Message processing - server](#) 18
[Messages](#)
[common data types](#) 14
[data types](#) 14
[transport](#) 14
[Methods](#)
[IAppHostAdminManager](#) 18
[IAppHostChangeHandler](#) 24
[IAppHostChildElementCollection](#) 25

- [IAppHostCollectionSchema](#) 27
- [IAppHostConfigException](#) 31
- [IAppHostConfigFile](#) 35
- [IAppHostConfigLocation](#) 40
- [IAppHostConfigLocationCollection](#) 44
- [IAppHostConfigManager](#) 47
- [IAppHostConstantValue](#) 49
- [IAppHostConstantValueCollection](#) 50
- [IAppHostElement](#) 52
- [IAppHostElementCollection](#) 61
- [IAppHostElementSchema](#) 66
- [IAppHostElementSchemaCollection](#) 70
- [IAppHostMappingExtension](#) 72
- [IAppHostMethod](#) 75
- [IAppHostMethodCollection](#) 77
- [IAppHostMethodInstance](#) 79
- [IAppHostMethodSchema](#) 82
- [IAppHostPathMapper](#) 84
- [IAppHostProperty](#) 85
- [IAppHostPropertyCollection](#) 91
- [IAppHostPropertyException](#) 93
- [IAppHostPropertySchema](#) 95
- [IAppHostPropertySchemaCollection](#) 103
- [IAppHostSectionDefinition](#) 105
- [IAppHostSectionDefinitionCollection](#) 110
- [IAppHostSectionGroup](#) 113
- [IAppHostWritableAdminManager](#) 118
- [Methods method](#) 60

N

Name method ([section 3.1.4.10.1](#) 49, [section 3.1.4.12.1](#) 52, [section 3.1.4.14.1](#) 66, [section 3.1.4.17.1](#) 76, [section 3.1.4.20.1](#) 82, [section 3.1.4.22.1](#) 86, [section 3.1.4.25.1](#) 95, [section 3.1.4.27.1](#) 105, [section 3.1.4.29.6](#) 116)

[Normative references](#) 11

O

[OnSectionChanges method](#) 25

[Output method](#) 79

[OutputSchema method](#) 83

OverrideModeDefault method ([section 3.1.4.27.4](#) 107, [section 3.1.4.27.5](#) 108)

[Overview \(synopsis\)](#) 11

P

[Parameters - security index](#) 125

[Path method](#) 41

[PossibleValues method](#) 101

[PostErrorLine method](#) 34

[Preconditions](#) 11

[PreErrorLine method](#) 33

[Prerequisites](#) 11

[Product behavior](#) 140

[Properties method](#) 53

[PropertySchemas method](#) 69

Protocol Details

- [overview](#) 15

R

[Read schema information: determine if ismergeappend is set in the defaultdocuments section example](#) 123

[Reading schema information example](#) 123

[References](#) 10

- [informative](#) 11
- [normative](#) 11

[Relationship to other protocols](#) 11

[Remove an entry of a collection example](#) 122

[RemoveElementSchema method](#) 29

[Removing entry of collection - example](#) 122

[RootSectionGroup method](#) 39

S

Schema method ([section 3.1.4.12.7](#) 58, [section 3.1.4.13.7](#) 65, [section 3.1.4.17.2](#) 76, [section 3.1.4.22.9](#) 91)

[Sections method](#) 115

Security

- [implementer considerations](#) 125
- [parameter index](#) 125

Sequencing rules

- [server](#) 18

[Sequencing rules - server](#) 18

Server

- [abstract data model](#) 15
- [IAppHostAdminManager method](#) 18
- [IAppHostChangeHandler method](#) 24
- [IAppHostChildElementCollection method](#) 25
- [IAppHostCollectionSchema method](#) 27
- [IAppHostConfigException method](#) 31
- [IAppHostConfigFile method](#) 35
- [IAppHostConfigLocation method](#) 40
- [IAppHostConfigLocationCollection method](#) 44
- [IAppHostConfigManager method](#) 47
- [IAppHostConstantValue method](#) 49
- [IAppHostConstantValueCollection method](#) 50
- [IAppHostElement method](#) 52
- [IAppHostElementCollection method](#) 61
- [IAppHostElementSchema method](#) 66
- [IAppHostElementSchemaCollection method](#) 70
- [IAppHostMappingExtension method](#) 72
- [IAppHostMethod method](#) 75
- [IAppHostMethodCollection method](#) 77
- [IAppHostMethodInstance method](#) 79
- [IAppHostMethodSchema method](#) 82
- [IAppHostPathMapper method](#) 84
- [IAppHostProperty method](#) 85
- [IAppHostPropertyCollection method](#) 91
- [IAppHostPropertyException method](#) 93
- [IAppHostPropertySchema method](#) 95
- [IAppHostPropertySchemaCollection method](#) 103
- [IAppHostSectionDefinition method](#) 105
- [IAppHostSectionDefinitionCollection method](#) 110
- [IAppHostSectionGroup method](#) 113
- [IAppHostWritableAdminManager method](#) 118
- [initialization](#) 18
- [local events](#) 120
- [message processing](#) 18
- [sequencing rules](#) 18
- [timer events](#) 120
- [timers](#) 18

[Set metadata: set the overridemode of the defaultdocument section example](#) 121

[SetMetadata method \(section 3.1.4.1.3 22, section 3.1.4.6.6 39, section 3.1.4.12.6 57, section 3.1.4.19.5 81, section 3.1.4.22.8 90\)](#)
[Setting metadata example](#) 121
[Standards assignments](#) 12
[StringValue method](#) 88

T

Timer events
 [server](#) 120
[Timer events - server](#) 120
Timers
 [server](#) 18
[Timers - server](#) 18
[TimeSpanFormat method](#) 103
[Tracking changes](#) 141
[Transport](#) 14
Type method ([section 3.1.4.25.2 96, section 3.1.4.27.2 106, section 3.1.4.27.3 106, section 3.1.4.29.7 117, section 3.1.4.29.8 117](#))

V

[ValidationFailureParameters method](#) 94
[ValidationFailureReason method](#) 94
[ValidationParameter method](#) 99
[ValidationType method](#) 99
Value method ([section 3.1.4.10.2 50, section 3.1.4.22.2 86, section 3.1.4.22.3 87](#))
[Vendor-extensible fields](#) 12
[Versioning](#) 11

W

[Write into a collection: add an entry for the defaultdocument section for site1 as a location tag example](#) 124
[Write into a collection: clear the contents of the defaultdocument section for site1 example](#) 123
Writing into collection example
 [adding entry as location tag](#) 124
 [clearing contents](#) 123